

GE Healthcare

Magmon3

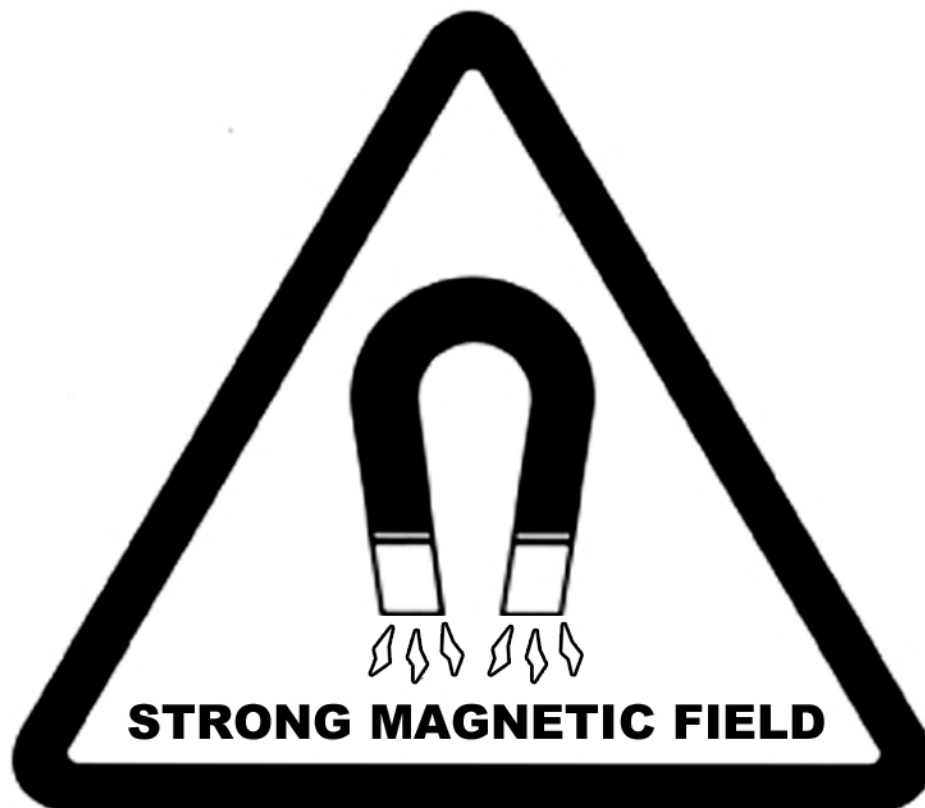
Installation & Service



OPERATING DOCUMENTATION

5124576
Revision 10

WARNING



NO PACEMAKERS NO METALLIC IMPLANTS

Persons with pacemakers, neurostimulators or metallic implants must not enter the magnet area. Serious injury may result.



NO LOOSE METAL OBJECTS

Iron and steel materials must not be taken into the magnet area. Serious injury or property damage may result.

Important Information

LANGUAGE

ПРЕДУПРЕЖДЕНИЕ (BG)

Това упътване за работа е налично само на английски език.

- Ако доставчикът на услугата на клиента изиска друг език, задължение на клиента е да осигури превод.
- Не използвайте оборудването, преди да сте се консултирали и разбрали упътването за работа.
- Неспазването на това предупреждение може да доведе до нараняване на доставчика на услугата, оператора или пациента в резултат на токов удар, механична или друга опасност.

警告 (ZH-CN)

本维修手册仅提供英文版本。

- 如果客户的维修服务人员需要非英文版本，则客户需自行提供翻译服务。
- 未详细阅读和完全理解本维修手册之前，不得进行维修。
- 忽略本警告可能对维修服务人员、操作人员或患者造成电击、机械伤害或其他形式的伤害。

警告 (ZH-HK)

本服務手冊僅提供英文版本。

- 倘若客戶的服務供應商需要英文以外之服務手冊，客戶有責任提供翻譯服務。
- 除非已參閱本服務手冊及明白其內容，否則切勿嘗試維修設備。
- 不遵從本警告或會令服務供應商、網絡供應商或病人受到觸電、機械性或其他危險。

警告 (ZH-TW)

本維修手冊僅有英文版。

- 若客戶的維修廠商需要英文版以外的語言，應由客戶自行提供翻譯服務。
- 請勿試圖維修本設備，除非 您已查閱並瞭解本維修手冊。
- 若未留意本警告，可能導致維修廠商、操作員或病患因觸電、機械或其他危險而受傷。

UPOZORENJE (HR)

Ovaj servisni priručnik dostupan je na engleskom jeziku.

- Ako davatelj usluge klijenta treba neki drugi jezik, klijent je dužan osigurati prijevod.
- Ne pokušavajte servisirati opremu ako niste u potpunosti pročitali i razumjeli ovaj servisni priručnik.
- Zanimarite li ovo upozorenje, može doći do ozljede davatelja usluge, operatera ili pacijenta uslijed strujnog udara, mehaničkih ili drugih rizika.

**VÝSTRAHA
(CS)**

Tento provozní návod existuje pouze v anglickém jazyce.

- V případě, že externí služba zákazníkům potřebuje návod v jiném jazyce, je zajištění překladu do odpovídajícího jazyka úkolem zákazníka.
- Nesnažte se o údržbu tohoto zařízení, aniž byste si přečetli tento provozní návod a pochopili jeho obsah.
- V případě nedodržování této výstrahy může dojít k poranění pracovníka prodejního servisu, obslužného personálu nebo pacientů vlivem elektrického proudu, respektive vlivem mechanických či jiných rizik.

**ADVARSEL
(DA)**

Denne servicemanual findes kun på engelsk.

- Hvis en kundes tekniker har brug for et andet sprog end engelsk, er det kundens ansvar at sørge for oversættelse.
- Forsøg ikke at servicere udstyret uden at læse og forstå denne servicemanual.
- Manglende overholdelse af denne advarsel kan medføre skade på grund af elektrisk stød, mekanisk eller anden fare for teknikeren, operatøren eller patienten.

**WAARSCHUWING
(NL)**

Deze onderhoudshandleiding is enkel in het Engels verkrijgbaar.

- Als het onderhoudspersoneel een andere taal vereist, dan is de klant verantwoordelijk voor de vertaling ervan.
- Probeer de apparatuur niet te onderhouden alvorens deze onderhoudshandleiding werd geraadpleegd en begrepen is.
- Indien deze waarschuwing niet wordt opgevolgd, zou het onderhoudspersoneel, de operator of een patiënt gewond kunnen raken als gevolg van een elektrische schok, mechanische of andere gevaren.

**WARNING
(EN)**

This service manual is available in English only.

- If a customer's service provider requires a language other than English, it is the customer's responsibility to provide translation services.
- Do not attempt to service the equipment unless this service manual has been consulted and is understood.
- Failure to heed this warning may result in injury to the service provider, operator or patient from electric shock, mechanical or other hazards.

**HOIATUS
(ET)**

See teenindusjuhend on saadaval ainult inglise keeles.

- Kui klienditeeninduse osutaja nõuab juhendit inglise keelest erinevas keeles, vastutab klient tõlketeenuse osutamise eest.
- Ärge üritage seadmeid teenindada enne eelnevalt käesoleva teenindusjuhendiga tutvumist ja sellest aru saamist.
- Käesoleva hoiatuse eiramine võib põhjustada teenuseosutaja, operaatori või patsiendi vigastamist elektrilöögi, mehaanilise või muu ohu tagajärjel.

**VAROITUS
(FI)**

Tämä huolto-ohje on saatavilla vain englanniksi.

- Jos asiakkaan huoltohenkilöstö vaatii muuta kuin englanninkielistä materiaalia, tarvittavan käännöksen hankkiminen on asiakkaan vastuulla.
- Älä yritä korjata laitteistoa ennen kuin olet varmasti lukenut ja ymmärtänyt tämän huolto-ohjeen.
- Mikäli tätä varoitusta ei noudateta, seurauksena voi olla huoltohenkilöstön, laitteiston käyttäjän tai potilaan vahingoittuminen sähköiskun, mekaanisen vian tai muun vaaratilanteen vuoksi.

**ATTENTION
(FR)**

Ce manuel d'installation et de maintenance est disponible uniquement en anglais.

- Si le technicien d'un client a besoin de ce manuel dans une langue autre que l'anglais, il incombe au client de le faire traduire.
- Ne pas tenter d'intervenir sur les équipements tant que ce manuel d'installation et de maintenance n'a pas été consulté et compris.
- Le non-respect de cet avertissement peut entraîner chez le technicien, l'opérateur ou le patient des blessures dues à des dangers électriques, mécaniques ou autres.

**WARNUNG
(DE)**

Diese Serviceanleitung existiert nur in englischer Sprache.

- Falls ein fremder Kundendienst eine andere Sprache benötigt, ist es Aufgabe des Kunden für eine entsprechende Übersetzung zu sorgen.
- Versuchen Sie nicht diese Anlage zu warten, ohne diese Serviceanleitung gelesen und verstanden zu haben.
- Wird diese Warnung nicht beachtet, so kann es zu Verletzungen des Kundendienst-technikers, des Bedieners oder des Patienten durch Stromschläge, mechanische oder sonstige Gefahren kommen.

**ΠΡΟΕΙΔΟΠΟΙΗΣΗ
(EL)**

Το παρόν εγχειρίδιο σέρβις διατίθεται μόνο στα αγγλικά.

- Εάν ο τεχνικός σέρβις ενός πελάτη απαιτεί το παρόν εγχειρίδιο σε γλώσσα εκτός των αγγλικών, αποτελεί ευθύνη του πελάτη να παρέχει τις υπηρεσίες μετάφρασης.
- Μην επιχειρήσετε την εκτέλεση εργασιών σέρβις στον εξοπλισμό αν δεν έχετε συμβουλευτεί και κατανοήσει το παρόν εγχειρίδιο σέρβις.
- Αν δεν προσέξετε την προειδοποίηση αυτή, ενδέχεται να προκληθεί τραυματισμός στον τεχνικό σέρβις, στο χειριστή ή στον ασθενή από ηλεκτροπληξία, μηχανικούς ή άλλους κινδύνους.

**FIGYELMEZTETÉS
(HU)**

Ezen karbantartási kézikönyv kizárólag angol nyelven érhető el.

- Ha a vevő szolgáltatója angoltól eltérő nyelvre tart igényt, akkor a vevő felelőssége a fordítás elkészítése.
- Ne próbálja elkezdni használni a berendezést, amíg a karbantartási kézikönyvben leírtakat nem értelmezték.
- Ezen figyelmeztetés figyelmen kívül hagyása a szolgáltató, működtető vagy a beteg áramütés, mechanikai vagy egyéb veszélyhelyzet miatti sérülését eredményezheti.

**ÁÐVÖRUN
(IS)**

Þessi þjónustuhandbók er aðeins fánleg á ensku.

- Ef að þjónustuveitandi viðskiptamanns þarfnast annas tungumáls en ensku, er það skylda viðskiptamanns að skaffa tungumálþjónustu.
- Reynið ekki að afgreiða tækið nema að þessi þjónustuhandbók hefur verið skoðuð og skilin.
- Brot á sinna þessari aðvörun getur leitt til meiðsla á þjónustuveitanda, stjórnanda eða sjúklings frá raflosti, vélrænu eða öðrum áhættum.

**AVVERTENZA
(IT)**

Il presente manuale di manutenzione è disponibile soltanto in lingua inglese.

- Se un addetto alla manutenzione richiede il manuale in una lingua diversa, il cliente è tenuto a provvedere direttamente alla traduzione.
- Procedere alla manutenzione dell'apparecchiatura solo dopo aver consultato il presente manuale ed averne compreso il contenuto.
- Il mancato rispetto della presente avvertenza potrebbe causare lesioni all'addetto alla manutenzione, all'operatore o ai pazienti provocate da scosse elettriche, urti meccanici o altri rischi.

**警告
(JA)**

このサービスマニュアルには英語版しかありません。

- サービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
- このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないでください。
- この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

**경고
(KO)**

본 서비스 매뉴얼은 영어로만 이용하실 수 있습니다.

- 고객의 서비스 제공자가 영어 이외의 언어를 요구할 경우, 번역 서비스를 제공하는 것은 고객의 책임입니다.
- 본 서비스 매뉴얼을 참조하여 숙지하지 않은 이상 해당 장비를 수리하려고 시도하지 마십시오.
- 본 경고 사항에 유의하지 않으면 전기 쇼크, 기계적 위험, 또는 기타 위험으로 인해 서비스 제공자, 사용자 또는 환자에게 부상을 입힐 수 있습니다.

**BRĪDINĀJUMS
(LV)**

Šī apkopes rokasgrāmata ir pieejama tikai angļu valodā.

- Ja klienta apkopes sniedzējam nepieciešama informācija citā valodā, klienta pienākums ir nodrošināt tulkojumu.
- Neveiciet aprīkojuma apkopi bez apkopes rokasgrāmatas izlasīšanas un saprašanas.
- Šī brīdinājuma neievērošanas rezultātā var rasties elektriskās strāvas trieciena, mehānisku vai citu faktoru izraisītu traumų risks apkopes sniedzējam, operatoram vai pacientam.

**ĮSPĖJIMAS
(LT)**

Šis eksploatavimo vadovas yra tik anglų kalba.

- Jei kliento paslaugų tiekėjas reikalauja vadovo kita kalba – ne anglų, suteikti vertimo paslaugas privalo klientas.
- Nemėginkite atlikti įrangos techninės priežiūros, jei neperskaitėte ar nesupratote šio eksploatavimo vadovo.
- Jei nepaisysite šio įspėjimo, galimi paslaugų tiekėjo, operatoriaus ar paciento sužalojimai dėl elektros šoko, mechaninių ar kitų pavojų.

**ADVARSEL
(NO)**

Denne servicehåndboken finnes bare på engelsk.

- Hvis kundens serviceleverandør har bruk for et annet språk, er det kundens ansvar å sørge for oversettelse.
- Ikke forsøk å reparere utstyret uten at denne servicehåndboken er lest og forstått.
- Manglende hensyn til denne advarselen kan føre til at serviceleverandøren, operatøren eller pasienten skades på grunn av elektrisk støt, mekaniske eller andre farer.

**OSTRZEŻENIE
(PL)**

Niniejszy podręcznik serwisowy dostępny jest jedynie w języku angielskim.

- Jeśli serwisant klienta wymaga języka innego niż angielski, zapewnienie usługi tłumaczenia jest obowiązkiem klienta.
- Nie próbować serwisować urządzenia bez zapoznania się z niniejszym podręcznikiem serwisowym i zrozumienia go.
- Niezastosowanie się do tego ostrzeżenia może doprowadzić do obrażeń serwisanta, operatora lub pacjenta w wyniku porażenia prądem elektrycznym, zagrożenia mechanicznego bądź innego.

**ATENÇÃO
(PT-BR)**

Este manual de assistência técnica encontra-se disponível unicamente em inglês.

- Se outro serviço de assistência técnica solicitar a tradução deste manual, caberá ao cliente fornecer os serviços de tradução.
- Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica.
- A não observância deste aviso pode ocasionar ferimentos no técnico, operador ou paciente decorrentes de choques elétricos, mecânicos ou outros.

**ATENÇÃO
(PT-PT)**

Este manual de assistência técnica só se encontra disponível em inglês.

- Se qualquer outro serviço de assistência técnica solicitar este manual noutro idioma, é da responsabilidade do cliente fornecer os serviços de tradução.
- Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica.
- O não cumprimento deste aviso pode colocar em perigo a segurança do técnico, do operador ou do paciente devido a choques eléctricos, mecânicos ou outros.

**ATENȚIE
(RO)**

Acest manual de service este disponibil doar în limba engleză.

- Dacă un furnizor de servicii pentru clienți necesită o altă limbă decât cea engleză, este de datoria clientului să furnizeze o traducere.
- Nu încercați să reparați echipamentul decât ulterior consultării și înțelegerii acestui manual de service.
- Ignorarea acestui avertisment ar putea duce la rănirea depanatorului, operatorului sau pacientului în urma pericolelor de electrocutare, mecanice sau de altă natură.

**ОСТОРОЖНО!
(RU)**

Данное руководство по техническому обслуживанию представлено только на английском языке.

- Если сервисному персоналу клиента необходимо руководство не на английском, а на каком-то другом языке, клиенту следует самостоятельно обеспечить перевод.
- Перед техническим обслуживанием оборудования обязательно обратитесь к данному руководству и поймите изложенные в нем сведения.
- Несоблюдение требований данного предупреждения может привести к тому, что специалист по техобслуживанию, оператор или пациент получит удар электрическим током, механическую травму или другое повреждение.

**UPOZORENJE
(SR)**

Ovo servisno uputstvo je dostupno samo na engleskom jeziku.

- Ako klijentov serviser zahteva neki drugi jezik, klijent je dužan da obezbedi prevodi-lačke usluge.
- Ne pokušavajte da opravite uređaj ako niste pročitali i razumeli ovo servisno uputstvo.
- Zanemarivanje ovog upozorenja može dovesti do povređivanja serviser, rukovaoca ili pacijenta usled strujnog udara ili mehaničkih i drugih opasnosti.

**UPOZORNENIE
(SK)**

Tento návod na obsluhu je k dispozícii len v angličtine.

- Ak zákazník poskytovateľ služieb vyžaduje iný jazyk ako angličtinu, poskytnutie prekladateľských služieb je zodpovednosťou zákazníka.
- Nepokúšajte sa o obsluhu zariadenia, kým si neprečítate návod na obsluhu a neporozumiete mu.
- Zanedbanie tohto upozornenia môže spôsobiť zranenie poskytovateľa služieb, obsluhujúcej osoby alebo pacienta elektrickým prúdom, mechanické alebo iné ohrozenie.

**ATENCION
(ES)**

Este manual de servicio sólo existe en inglés.

- Si el encargado de mantenimiento de un cliente necesita un idioma que no sea el inglés, el cliente deberá encargarse de la traducción del manual.
- No se deberá dar servicio técnico al equipo, sin haber consultado y comprendido este manual de servicio.
- La no observancia del presente aviso puede dar lugar a que el proveedor de servicios, el operador o el paciente sufran lesiones provocadas por causas eléctricas, mecánicas o de otra naturaleza.

**VARNING
(SV)**

Den här servicehandboken finns bara tillgänglig på engelska.

- Om en kunds servicetekniker har behov av ett annat språk än engelska, ansvarar kunden för att tillhandahålla översättningstjänster.
- Försök inte utföra service på utrustningen om du inte har läst och förstår den här servicehandboken.
- Om du inte tar hänsyn till den här varningen kan det resultera i skador på serviceteknikern, operatören eller patienten till följd av elektriska stötar, mekaniska faror eller andra faror.

**OPOZORILO
(SL)**

Ta servisni priročnik je na voljo samo v angleškem jeziku.

- Če ponudnik storitve stranke potrebuje priročnik v drugem jeziku, mora stranka zagotoviti prevod.
- Ne poskušajte servisirati opreme, če tega priročnika niste v celoti prebrali in razumeli.
- Če tega opozorila ne upoštevate, se lahko zaradi električnega udara, mehanskih ali drugih nevarnosti poškoduje ponudnik storitev, operater ali bolnik.

**DİKKAT
(TR)**

Bu servis kılavuzunun sadece ingilizcesi mevcuttur.

- Eğer müşteri teknisyeni bu kılavuzu ingilizce dışında bir başka lisandan talep ederse, bunu tercüme ettirmek müşteriye düşer.
- Servis kılavuzunu okuyup anlamadan ekipmanlara müdahale etmeyiniz.
- Bu uyarıya uyulmaması, elektrik, mekanik veya diğer tehlikelerden dolayı teknisyen, operatör veya hastanın yaralanmasına yol açabilir.

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Revision History

Revision	Date	Description
1	May 11, 2005	Initial release.
2	June 15, 2005	Changes throughout based on pilot feedback.
3	July 18, 2005	Changes throughout based on feedback from HII Review.
4	June 22, 2006	Reworked Installation order for clarity and added basic setup procedure.
5	Dec. 21, 2006	Updated entire manual for flow and function.
6	May 9, 2008	Updated Configuration Setup for the new compressor types. Per PQR 13128295: Changed title of Magmon3 Non-LCC Cable Collector to "Magmon3 Non-Zero Boil-Off Cable Collector" and updated chapter on Magmon3 Installation.
7	Jan. 15, 2009	Per PQR 13128295: Updated chapter on Magmon3 Installation with references to LX, Excite and a section for installing the MR750 Heat Exchanger Cabinet. Per PQR 13224885: Updated the LCC 1.0T/1.5T Mobile Van interconnect diagram in the chapter, Interconnect Maps.
8	June 17, 2009	Per PQR 13223043: Updated the chapter on Magmon3 Installation by adding pan-head screws to Section 2.2 Required Tools and Equipment. Per PQR 13211352: Updated Leybold and Sumitomo compressor models in Magmon3 Installation.
9	Nov. 15, 2009	Per SCR DOC0680558: Updated Magmon3 Front Panel User Interface, Magmon3 Installation, Magmon3 Installation & Service, and Magmon3 Web Page Screen Overview.
10	Oct. 25, 2010	Per SPR MRIhc48551: Profiled text and sections in chapters appropriately for Class A version of the manual. Updated Table 2-10. ECO 2089099: Reed switch removed per VCP.

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Chapter 1 Specifications

1 Magmon3 Component Specifications

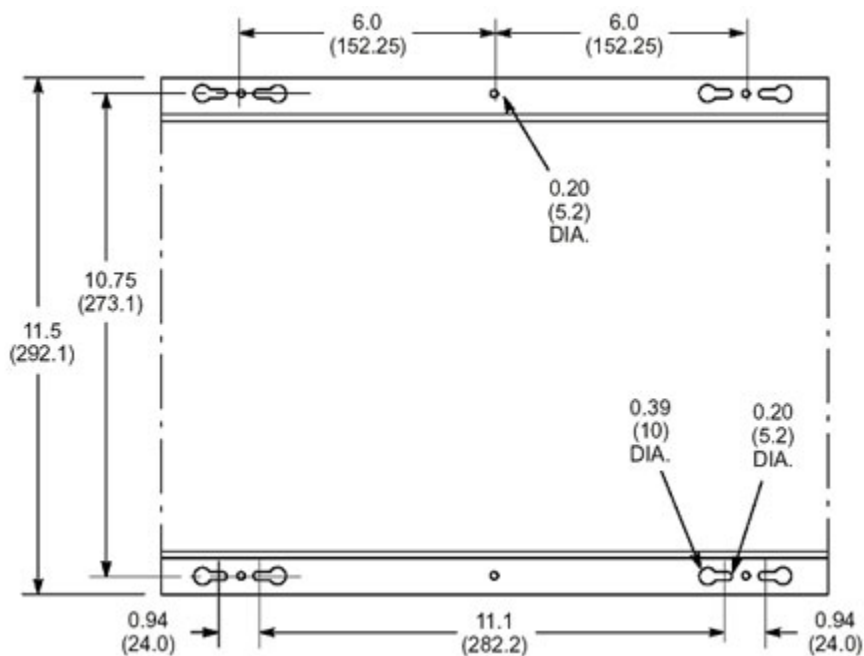
1.1 Magmon3 – Part Number 2394952

1.1.1 Mechanical Specifications

Table 1-1: Magmon3 Mechanical Specifications

Parameter	Specification
Dimensions	Height: 10.0 in (254 mm) Width: 15.0 in (381 mm) Depth: 3.0 in (76 mm)
Weight	8 pounds (3.6 kg)
Mounting	Wall mount; For systems with GE-supplied MDP, this unit mounts next to it. NOTE: This unit should not be mounted in the exhaust path of any equipment cabinet.
Service Clearance	Maintain 12-inch (305 mm) clearance above, below and on both sides of the unit.

Illustration 1-1: Magmon3 Dimensions



NOTE: All dimensions in this diagram are in inches. All bracketed () dimensions are in millimeters.

1.1.2 Environmental Specifications

Table 1-2: Magmon3 Environmental Specifications

Parameter	Specification
Operating Temperature Non-Operating Temperature	0 to 40 Degrees C (32 to 104 Degrees F) -34 to 50 Degrees C (-29 to 122 Degrees F)
Temperature Change	5 Degrees C / Hr
Operating Relative Humidity Non-Operating Relative Humidity	10 to 80% (non-condensing) 10 to 90% (non-condensing)
Magnetic Field	< 100 gauss
Altitude	-30 meters to 2133 meters; above sea level

1.1.3 Electrical Specifications

Table 1-3: Magmon3 Electrical Specifications

Parameter	Specification
Input Voltage	100 – 120 Vac or 200 – 220 Vac (auto switching)
Line Frequency	50 or 60 Hz (auto switching)
Nominal Current	0.5 amperes

1.2 Water Flow / Temperature Meter (Turbine Type) – Part Number 2333825

1.2.1 Mechanical Specifications

Table 1-4: Water Flow / Temperature Meter Mechanical Specifications

Parameter	Specification
Dimensions	Height: 2.75 in (69.9 mm) Length: 3.70 in (94 mm) Depth: 2.0 in (50.8 mm)
Weight	2.75 pounds (1.3 kg)
Mounting	Horizontal on any vertical surface. NOTE: Must not lay on the floor, this will result in performance issues.
Maximum Fluid Pressure	200 psi
Maximum Fluid Temperature	225 Degrees F (107.2 Degrees C)
Inlet / Outlet Piping	½-inch NPT × 5-inch long brass pipe. NOTE: Straight pipe required for non-turbulent laminar flow.
Non-Wetted Materials Composition	Epoxy, Lexan, and PVC
Wetted Materials Composition	316 stainless steel, Buna-N, Delrin, Acetal Copolymer
Flow Direction	Operates with flow from either direction.
Mounting	Horizontal on any vertical surface with piping located at top. See Illustration 1-2 . NOTE: Must not lay on the floor, this will result in performance issues.

Illustration 1-2: Flow Meter Mounting Orientation



1.2.2 Environmental Specifications

Table 1-5: Water Flow / Temperature Meter Environmental Specifications

Parameter	Specification
Operating Temperature Non-Operating Temperature	0 to 40 Degrees C (32 to 104 Degrees F) -34 to 50 Degrees C (-29 to 122 Degrees F)
Temperature Change	5 Degrees C / Hr
Operating Relative Humidity Non-Operating Relative Humidity	10 to 80% (non-condensing) 10 to 90% (non-condensing)
Magnetic Field	< 10 gauss
Altitude	-30 meters to 2133 meters; above sea level

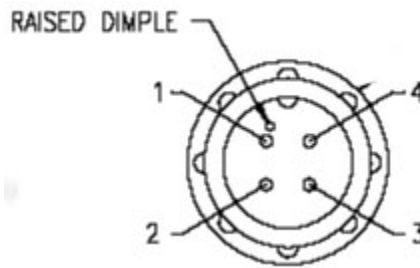
1.2.3 Electrical Specifications

Table 1-6: Water Flow / Temperature Meter Electrical Specifications

Parameter	Specification
Input Voltage	+12 to 35 Vdc, < 25 ma
Transmission Distance	< 200 feet (6096 cm)
Output Voltage (analog)	0 to 5 Vdc, for both flow and temperature
Calibrated Flow Range	0 to 5 gpm (0 to 18.9 lpm)
Calibrated Temperature Range	0 to 100 Degrees F (0 to 38.8 Degrees C)
Accuracy	± 2% of Range
Repeatability	0.5% of Range
Resolution	Infinite
Response Time (Flow Time)	2 seconds to 90% step change
Calibration	Factory-calibrated, no service calibrations required

1.2.4 Electrical Connections

Illustration 1-3: Electrical Connections



Pin 1: + Positive DC Supply

Pin 2: Common Ground

Pin 3: Flow Sensor Analog Output

Pin 4: Temperature Sensor Analog Output

1.3 Pressure Transducer

1.3.1 Pressure Transducer – Part Number 2299843 and 2299843-2

NOTE: Out of Catalog Item – Manufacturer Name: SUNX

The manufacturer is responsible for the design and specifications of this unit. Any changes to specs can be viewed on their website: <http://www2.sunx.co.jp/english/productindex.jsp> (search for model DP2-41N).

1.3.1.1 Mechanical Specifications

Table 1-7: Pressure Transducer Mechanical Specifications

Parameter	Specification
Application	For Air or Non-corrosive gases
Dimensions	Width: 1.181 in (30 mm) Height: 1.760 in (44.7 mm) Depth: 1.378 in (35 mm)
Weight	4.2 ounces (120 grams)
Pressure Port	1/8-inch Female NPT (port orientation defined by GE part number)
Pressure	Positive Pressure Measurement
Rated Pressure Range	0 to 14.5 psi (0 to 100 kPa)
Resolution	0.015 psi (0.1 kPa)
Max Pressure	71 psi (490 kPa)

1.3.1.2 Environmental Specifications

Table 1-8: Pressure Transducer Environmental Specifications

Parameter	Specification
Operating Temperature	-10 to 50 Degrees C (32 to 104 Degrees F)
Non-Operating Temperature	-10 to 60 Degrees C (-29 to 122 Degrees F)
Operating Relative Humidity	35 to 85% (non-condensing)
Non-Operating Relative Humidity	35 to 85% (non-condensing)

1.3.1.3 Electrical Specifications

Table 1-9: Pressure Transducer Electrical Specifications

Parameter	Specification
Input Voltage	12 to 24 Vdc + 10% and – 15%
Ripple P-P	< 10 %
Analog Voltage Output	1 to 5 Vdc (NPN open collector transistor)
Response Time	< 2.5 milliseconds
Short Circuit Protection	Incorporated

1.3.1.4 Product Features

Table 1-10: Pressure Transducer Product Features

Feature	Description
3.5-Digit Red LED Display	0.394 inch (10 mm) high characters The unit displays measured pressure, settings, error messages and key-protected status.
Displayable Pressure Range	-0.725 to 14.5 psi (-5.0 to 100 kPa)
Sample Rate	4 times / second
Analog Bar Display	LED Bar Display in steps of 10% full scale
Operation Indicators	
Comparative Output 1	Orange LED illuminates when Output 1 is ON
Comparative Output 2	Green LED illuminates when Output 2 is ON

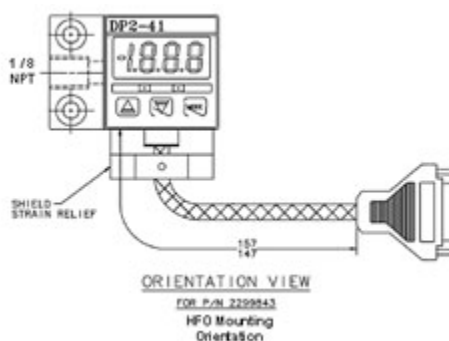
1.3.1.5 Electrical Connections

Table 1-11: Pressure Transducer Electrical Connections – DB-9 Female

Pin Number	Signal Name
3	Common Ground
6	Analog Output Signal
8	+12 Vdc input

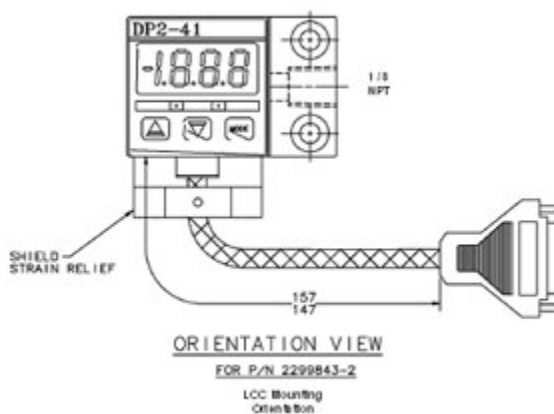
1.3.1.6 HFO Mounting Orientation

Illustration 1-4: HFO Mounting Orientation



1.3.1.7 LCC Mounting Orientation

Illustration 1-5: LCC Mounting Orientation



1.3.2 Pressure Transducer – Part Number 5192503

NOTE: The manufacturer is responsible for the specifications of this unit. Any changes to specs can be viewed on their website under model S-10: http://www.wika.us/upload/DS_PE_S10_en_us_16325.pdf

Illustration 1-6: Pressure Transducer



1.3.2.1 Mechanical Specifications

Table 1-12: Pressure Transducer Mechanical Specifications

Parameter	Specification
Application	Hydraulics and Pneumatics, Test Equipment

Dimensions	Case OD: 27 mm Height: 311 mm
Weight (minus cable)	0.4 lbs
Pressure Port	¼-inch NPT
Pressure	Positive Pressure Measurement
Rated Pressure Range	0 to 15 psi
Max Pressure	70 psi

1.3.2.2 Environmental and Electrical Specifications

Table 1-13: Pressure Transducer Environmental and Electrical Specifications

Parameter	Specification
Temperature	-40 - 125 C
Input Voltage	12 to 24 Vdc + 10% and – 15%
Analog Voltage Output	1 to 5 Vdc (NPN open collector transistor)
Response Time	< 2.5 milliseconds
Short Circuit Protection	Incorporated

1.3.2.3 Electrical Connections

Table 1-14: Pressure Transducer Electrical Connections – DB-9 Female

Pin Number	Signal Name
3	Common Ground
6	Analog Output Signal
8	+12 Vdc input

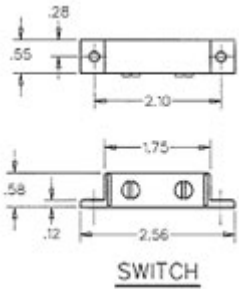
1.4 Reed Switch Part Number 2108790

The reed switch is used to detect the presence of the magnetic field on the MRI Scanner. The reed switch contacts are normally CLOSED when in the presence of a magnetic field. Loss of magnetic field or an unplugged cable will result in OPEN contacts.

NOTE: The magnet monitor does not initiate any alarms with the change in state of the reed switch. Magnet Monitor kits manufactured for the DV platform systems after March 2010 do not include the reed switch.

1.4.1 Mechanical Specifications

Table 1-15: Reed Switch Mechanical Specifications

Parameter	Specification
Dimensions	
Mounting	Attaches to rear pedestal with cable tie wraps.

1.4.2 Environmental Specifications

Table 1-16: Reed Switch Environmental Specifications

Parameter	Specification
Operating Temperature	10 to 70 Degrees C (50 to 158 Degrees F)
Non-Operating Temperature	-34 to 70 Degrees C (-29 to 158 Degrees F)
Operating Relative Humidity	10 to 80% (non-condensing)
Non-Operating Relative Humidity	10 to 90% (non-condensing)
Altitude	-30 meters to 2133 meters; above sea level

1.4.3 Electrical Specifications

Table 1-17: Reed Switch Electrical Specifications

Parameter	Specification
Switch Contacts – Maximum Voltage	30 Vdc
Switch Contacts – Maximum Current	0.25 Amps dc
Initial Resistance	.25 ohms
Voltage Hold off	200 Vdc
Watts – Maximum	3.0 W.
Contact Operation	SPST – Normally Closed

1.5 Remote Alarm Box – Part Number 2266628

1.5.1 Mechanical Specifications

The remote alarm box is used on HFO and 7.0T systems. It is optional for all other systems.

Table 1-18: Remote Alarm Box Mechanical Specifications

Parameter	Specification
Dimensions	Width: 3 in (76.2 mm) Length: 3 in (76.2 mm) Depth: 1.5 in (38.1 mm)
Mounting	Near the Operator Workspace Table; in view of the operator. Table top surface or wall mount surface with Velcro.

Illustration 1-7: Remote Alarm Box



1.5.2 Environmental Specifications

Table 1-19: Remote Alarm Box Environmental Specifications

Parameter	Specification
Operating Temperature	10 to 40 Degrees C (50 to 104 Degrees F)
Non-Operating Temperature	-34 to 50 Degrees C (-29 to 122 Degrees F)
Temperature Change	5 Degrees C / Hr
Operating Relative Humidity	10 to 80% (non-condensing)
Non-Operating Relative Humidity	10 to 90% (non-condensing)
Noise	< 50 dB
Magnetic Field	< 100 gauss
Altitude	-30 meters to 2133 meters; above sea level

1.5.3 Electrical Specifications

Table 1-20: Remote Alarm Box Electrical Specifications

Parameter	Specification
Input Voltage	12 Vdc $\pm$ 10 %
Transition Time	Rise time of 10 microseconds or less Fall time of 1 millisecond or less

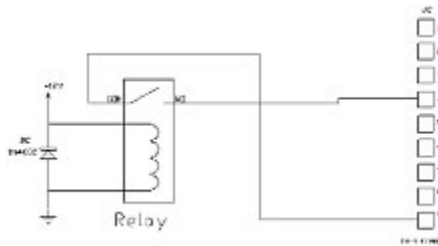
Parameter	Specification
Output Relay Contacts	1 ampere at a maximum 120 Vac rms

1.5.4 Electrical Connections

Table 1-21: Remote Alarm Box Electrical Connections

Pin	Signal
Input Connector J1; DB-9 Pin; Male	
2	12 Vdc
9	Common Ground
Output Connector J2; DB-9; Female	
4	Relay Contact (Normally Open Contact)
9	Relay Contact Common

Illustration 1-8: Output Relay and Connector Schematic



1.5.5 Functional Specifications

Table 1-22: Remote Alarm Box Functional Specifications

Device	Parameter	Specification
Audible Device	Audio Tone	Frequency Range 1000 to 3000 Hz
	Audible Level	50 dBA to 90 dBA; measured 1 meter from audible device.
	Pulsing Frequency	Pulse On / Off at a rate of 1 Hz $\pm$ 20%.
Visual Indicator	Type	LED
	Color	Red
	Intensity	Noticeable at 20 feet with a conical viewing angle of $\pm$ 5 degrees.
	Pulsing Frequency	Pulse On / Off at a rate of 1 Hz $\pm$ 20%.
	Reset Function	When the 12 Vdc input signal is removed, then the LED will be extinguished.
Output Relay	Non Alarm Mode	Contact closure when 12 Vdc is applied to input J1 pin 2
	Alarm Mode	N/O Contact when 12 Vdc is removed from input J1 pin 2
Silence Button	Alarm Mode	When depressed will silence the audible device when in alarm.

1.6 RUO Pre-Amplifier Part Number 2219341

The RUO Pre-Amplifier is a Dual Channel differential input, single ended output device. The RUO Pre-Amplifier also incorporates the necessary interconnections to route stimulus currents to the sensors located in the Coldhead and Magnet Sleeve.

Illustration 1-9: RUO Pre-Amplifier



1.6.1 Mechanical Specifications

Table 1-23: RUO Pre-Amplifier Mechanical Specifications

Parameter	Specification
Dimensions	Width: 2 in (50.8 mm) Length: 3.75 in (95.25 mm) Depth: 1.5 in (38.1 mm)
Weight	5 ounces (142 grams)
Mounting	Surface mount with Velcro; On top of the magnet near the Coldhead.

1.6.2 Environmental Specifications

Table 1-24: RUO Pre-Amplifier Environmental Specifications

Parameter	Specification
Operating Temperature Non-Operating Temperature	0 to 40 Degrees C (32 to 104 Degrees F) -34 to 50 Degrees C (-29 to 122 Degrees F)
Temperature Change	5 Degrees C / Hr
Operating Relative Humidity Non-Operating Relative Humidity	10 to 80% (non-condensing) 10 to 90% (non-condensing)
Noise	< 50 dB
Magnetic Field	< 1500 gauss
Altitude	-30 meters to 2133 meters; above sea level

1.6.3 Electrical Specifications

Table 1-25: RUO Pre-Amplifier Electrical Specifications

Parameter	Specification
Input Voltage	± 12 Vdc
Gain per Channel	$65 \pm 1\%$
Channel to Channel Isolation	≥ 60 dB
Output Noise	≤ 50 microvolts rms

Parameter	Specification
Bandwidth	3 dB bandwidth > 10 KHz

1.6.4 Connectors And Input / Output Signals

Table 1-26: Input Connector J1; To/From Sensors; Type DB15 pin; Female

Pin	Signal Description	Signal Level
1	Stage 1 Diode Positive	10 microamp constant current
2	Stage 1 Diode Negative	
3	Stage 2 Diode Positive	10 microamp constant current
4	Stage 2 Diode Negative	
5	RUO (1) Source Positive	10 microamp constant current
6	RUO (1) Source Negative	
7	RUO (1) Signal Positive	0 to $\pm$ 20 millivolts dc
8	RUO (1) Signal Negative	
9	No Connection	n/a
10	No Connection	n/a
11	No Connection	n/a
12	RUO (2) Source Positive	10 microamp constant current
13	RUO (2) Source Negative	
14	RUO (2) Signal Positive	0 to $\pm$ 20 millivolts dc
15	RUO (2) Signal Negative	

Table 1-27: Output Connector J2; To/From MagMon; Type DB15 pin; Male

Pin	Signal Description	Signal Level
1	Stage 1 Diode Positive	10 microamp constant current
2	Stage 1 Diode Negative	
3	Stage 2 Diode Positive	10 microamp constant current
4	Stage 2 Diode Negative	
5	RUO (1) Source Positive	10 microamp constant current
6	RUO (1) Source Negative	
7	RUO (1) Output	0 to $\pm$ 2.5 Vdc
8	Signal Ground	0 Vdc
9	No Connection	n/a
10	No Connection	n/a
11	No Connection	n/a
12	RUO (2) Source Positive	10 microamp constant current
13	RUO (2) Source Negative	
14	RUO (2) Output	0 to $\pm$ 2.5 Vdc

Pin	Signal Description	Signal Level
15	RUO (2) Signal Negative	0 Vdc

1.7 UPS Part Number 2276094

NOTE: Out of Catalog Item – Manufacturer: Powerware

The manufacturer is responsible for the design and specifications of this unit. Any changes to specs can be viewed on their website: <http://powerquality.eaton.com/Products-services/legacy/9120-info.asp>

1.7.1 Overview: Powerware 9120 UPS

The Powerware 9120 UPS provides maximum power quality and backup power protection in the 700 - 3000 VA range and is the ideal power management solution for networks, Web servers and telecommunications equipment.

While offering basic surge protection and backup power, the Powerware 9120 UPS additionally offers the best UPS power protection against all nine common power quality problems.

1.7.2 Product Snapshot

Table 1-28: UPS Product Snapshot

Parameter	Specification
Model Number	PW9120/700
Power Rating	700 VA
Voltage	120 and 230 V
Frequency	50 and 60 Hz auto sensing
Configuration	Tower
Input Connection	5-15P
Output Receptacles	(4) 5-15R
Dimensions	Height: 9.6 in (243 mm) Width: 6.2 in (158 mm) Depth: 16.2 in (412 mm)
Weight	29 lbs (13.2 kg)

1.7.3 Electrical Input

Table 1-29: UPS Electrical Input

Parameter	Specification
Nominal Voltage	120 Vac and 230 Vac
Input Voltage Range	120V: 80 –144 Vac 230V: 120/140/160-276 Vactd>
Input Power Factor	>.95%
Operating Frequency	50/60 Hz, Auto-sensing
Frequency Range	45 - 65 Hz

Parameter	Specification
Input Protection	Fuse or circuit breaker

1.7.4 Electrical Output

Table 1-30: UPS Electrical Output

Parameter	Specification
On Utility Voltage Regulation	±2% of nominal
On Battery Voltage Regulation	±3% of nominal
Nominal Output Voltage	Same as selected input voltage
Output Voltage Waveform	Sine Wave
Output Voltage Distortion	<3% THD
Output Protection	Electronic overload sensing, and circuit breaker protection
Efficiency	Online Mode: >86% Hi-Efficiency Mode: >90%

1.7.5 Battery

Table 1-31: UPS Battery

Parameter	Specification
Internal Battery Type	Sealed, lead-acid; maintenance free
EBM Battery Type	Sealed, lead-acid; maintenance free
On Battery Runtimes	See Powerware website: http://www.powerware.com/UPS/#battery
Battery Replacement	Hot-swappable internal and external batteries
Recharge Time	< 4 hours to 90% capacity
Start-On-Battery	Allows start of UPS without utility input

1.7.6 Communications

Table 1-32: UPS Communications

Parameter	Specification
User Interface	LCD Status screen
Audible Alarms	UPS alarm conditions, including: On-Battery, Low Battery, Overload, UPS Fault
Network Transient Protector	In and out jack for all models. UL497A tested.
REPO Port	Meets NEC Code 645-11 intent and UL requirements
Communications	1 RS232 Serial Port; 1 communications slot; 1 USB port
Communications Cable	6-foot communications cable included
Power Management Software	Powerware Software Suite CD (http://www.powerware.com/Software/Suite.asp)

1.7.7 General

Table 1-33: UPS General

Parameter	Specification
Topology	True online, double-conversion
Diagnostics	Full system self-test on power up
UPS Bypass	Automatic on Overload or UPS failure <4ms
Transfer Time to Battery	0 msq
Overload Capacity	125% for 10 minutes before transfer to bypass. 150% for 10 seconds before transfer to bypass.

1.7.8 Environmental and Safety

Table 1-34: UPS Environmental and Safety Specifications

Parameter	Specification
Safety Certifications	UL1778;cUL22.2 NO.107.1
EMI Compliance	FCC Part 15, Class B (700-1500), Class A (2000-3000) 230 V, EN 50091-2 Class B (700-1500), Class A (2000-3000)
Operating Temperature	0 Degrees to 40 Degrees C (32 to 104 F)
Storage Temperature	-15 Degrees to 50 Degrees C (5 to 122 F)
Storage Temperature	0 Degrees to 25 Degrees C (32 to 77 F)
Relative Humidity	0% to 95% non-condensing
Immunity	EEE C62.41, IEC 61000-4 -2, -3, -4, -5
Network Transient Protector	UL497A
Audible Noise at 1 meter	700-1000 VA: <45dB; 1500 VA <50dB; 2000-3000 <52dB
Altitude	3000m (10,000 ft) without deteriorating

NOTE: Due to continuing product improvement programs, specifications are subject to change without notice. See <http://powerquality.eaton.com/Products-services/legacy/9120-info.asp> for the latest information.

1.8 Global Modem Part Number 2245773

NOTE: Out of Catalog Item – Manufacturer Name: MultiTech
The manufacturer is responsible for the design and specifications of this unit.

These specifications are taken from the *MultiModem® ZBA User Guide for Models MT5634ZBA-V.90, MT5634ZBA-V-V.90, MT5634ZBA-V.92, MT5634ZBA-V-V.92.*

1.8.1 Mechanical Specifications

Table 1-35: Global Modem Mechanical Specifications

Parameter	Specification
Dimensions	Width: 4.25 in (10.8 cm) Length: 5.8 in (14.8 cm) Depth: 1.15 in (2.9 cm)
Weight	8 ounces (224 grams)
Mounting	Near the Magnet Monitor Unit. Table top surface or wall mount surface with Velcro.
Connectors	DB25F RS232C Connector; 2 RJ-11 Telephone jacks; power jack

1.8.2 Environmental Specifications

Table 1-36: Global Modem Environmental Specifications

Parameter	Specification
Temperature	0 to 50 Degrees C (32 to 122 Degrees F)
Relative Humidity	20 to 90% (non-condensing)
Magnetic Field	< 100 gauss
Altitude	-30 meters to 2133 meters; above sea level

1.8.3 Electrical Specifications

Table 1-37: Global Modem Electrical Specifications

Parameter	Specification
Line Voltage	100 to 240 Vac; universal power supply
Line Frequency	50/60 Hz; universal power supply
Power Consumption	9 Vdc; 300 milliamps maximum

1.8.4 Features

Table 1-38: Global Modem Features

Parameter	Specification
Indicators	LEDs for Transmit Data, Receive Data, Carrier Detect, 56k bps, 33.6k bps, 14.4k bps, Off Hook, Terminal Ready, Error Correction, FAX
Manual Control	Power Switch
Speaker	Internal Speaker for call progress monitoring
Diagnostics	Power-on self test, local analog loop, local digital loop, remote digital loop.

1.9 Satellite Modem Part Number 2387630

NOTE: Out of Catalog Item - Manufacturer: Stellar Satellite Communications

The manufacturer is responsible for the design and specifications of this unit. Any changes to specs can be viewed on their website.

<http://www.stellar-sat.com>

The ST2500 is a highly versatile smart modem, with low-power modes; extensive serial, digital and analog I/O; and an application processor capable of hosting customer-written "C" code. It is available with optional enclosure and 12-channel GPS.

1.9.1 Satellite Modem Specifications

Illustration 1-10: Stellar ST2500 Satellite Modem

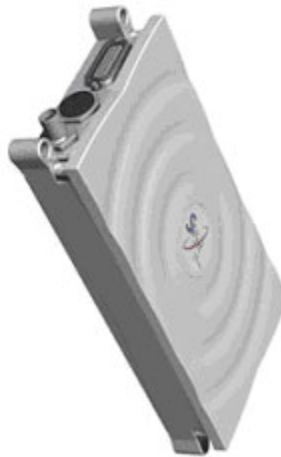
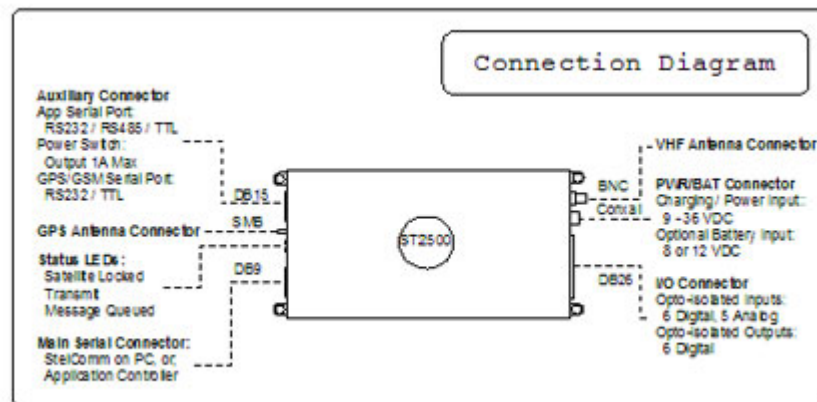


Table 1-39: Satellite Modem Specifications

Parameter	Specification
General	Dimension (OEM board): 90 × 150 × 32 mm Dimension (Case): 222 × 110 × 35 mm LED Indicators: Receive, Transmit, Queue ORBCOMM Hybrid Mode: Yes
Transmit Frequency	148 to 150.05 MHz
Transmit Power	5 Watts
Receive Frequency	137 to 138 MHz
Dynamic Range	40 dB minimum
Sensitivity	Minimum BER: E-5@ -118 dBm Typical BER: E-5@ -120 dBm
Application Interfaces	Main Serial: RS232 Aux1 Serial: RS232 to Sensor Aux2 Serial: TTL/RS232 to GPS, GSM Modem Digital Inputs: 6 optically isolated Digital Outputs: 6 optically isolated Analog Inputs: 5 (0 to 3V, 10 bit)
Application Programming	Application Processor: Atmel Mega 128L Program Memory: 128K Flash Data Memory: 4K RAM, 4K EEPROM Programming Language: C Compiler: IAR or Imagecraft
Power Requirements	External: 9 to 36 VDC Battery: 8 or 12 VDC
Power	Transmit: 2.5 A max.@12v; 2.0 A max.@15v Receive: 90 mA Power Save Receive: 60 mA Sleep: 50 uA typical, 100 uA maximum
Environmental	Operating: -40°C to +85°C Shock & Vibration: SAE J1455
Position Accuracy	GPS: < 5 m CEP (varies by GPS manufacturer)
Options	Trimble LP 8-channel GPS or Fastrax 12-channel GPS (requires 3.3V Antenna).

Illustration 1-11: Connection Diagram



<http://www.orbcomm.com>

For more information regarding ORBCOMM modems, contact the modem information line at 703-433-6520

1.9.2 MRI Mobile Installation

Illustration 1-12: MRI Mobile Installation



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Chapter 2 Product Structure

1 Magmon3 Product Structure Information

The hardware and cable collectors in [Table 2-1](#) and its related documents may be used for taking inventory of parts received prior to installation.

If any parts are found to be missing, submit a Missing In Shipment report to receive parts. Information can be found in the front material of the Installation Manual.

If any received part is found to be defective, then code as DOA and order a replacement.

Table 2-1: Magmon3 Top Level Shipping Collectors

Line	Collector Part Number	Description
1	2222520-3	<i>Magmon3 - LCC Fixed Site, Transportable, and Relocatable Collector</i> Used on Forward Production LCC 1.5T and LCC 300 Magnets Collector Contents 2394952: Magmon3, with power cord and laptop network crossover table 2219341: RUO Pre-Amp Box Magmon3 LCC Cable Collector, Part 2350809-2 Magmon3 Global Modem Collector, Part 2245794-2
2	2222520-4	<i>Magmon3 - LCC Mobile Collector</i> Used on Forward Production LCC 1.5T Magnet Collector Contents 2394952: Magmon3, with power cord and laptop network crossover table 2219341: RUO Pre-Amp Box Magmon3 LCC Cable Collector, Part 2350809-2 Mobile Satellite Modem Collector, Part 2385399 2276094: Uninterruptible Power Supply 2276569: Cable Run 939 - MDP-UPS Out to MSM4 Magmon3 Remote Alarm Collector, Part 5180301
3	2222520-5	<i>Magmon3 - Non-LCC Fixed Site, Transportable and Relocatable Collector [Non-LCC Magnet Types: S2 / S3 (including Magnishield) / S4 / S5 / 1.0T SX / 1.0T; 1.5T CX / 0.5T Max])</i> Used on HD upgrades and Installed Base GE magnets with no magnet monitor. Collector Contents 2394952: Magmon3, with power cord and laptop network crossover table Magmon3 Non-LCC Cable Collector, Part 2350809-3 Magmon3 Global Modem Collector, Part 2245794-2
4	2222520-6	<i>Magmon3 - HFO Fixed Site Collector</i> used on Forward Production 0.7T High Field Open Magnets Collector Contents 2394952: Magmon3, with power cord and laptop network crossover table 2219341: RUO Pre-Amp Box Magmon3 HFO Cable Collector, Part 2265809-3 Magmon3 Global Modem Collector, Part 2245794-2 Magmon3 Remote Alarm Collector, Part 5180301 2276094: Uninterruptible Power Supply

Line	Collector Part Number	Description
5	2404412	<i>Magmon3 - GEMSO 7.0T Fixed Site Collector</i> Used on Forward Production 7.0T GEHCO Magnets. Collector Contents 2394952: Magmon3, with power cord and laptop network crossover table 2219341: RUO Pre-Amp Box Magmon3 GEHCO 7.0T Cable Collector, Part 2350809-4 2333825: Water Flow (Turbine)/Temperature Sensor Assembly (2 each) 2266628: Magmon3 Remote Alarm Box 2404712: Lock Coil Timer 46-294231G1: MRU 2404413: Instrumentation Box
6	5267626	<i>Magmon3 - MR750/MR450/MR450W, Collector</i> Used on 3.0T and 1.5T Systems. Collector Contents 2394952: Magmon3, with power cord and laptop network crossover table 2219341: RUO Pre-Amp Box Magmon3 MR750/MR450 Cable Collector, Part 5193795

2 Collectors

2.1 Global Modem Collector: Part 2245794-2

Table 2-2: Global Modem Collector Contents

Part Number	Item
2245773	Multi-Tech Global Modem
2253088	Label, Customer Property for 2245794
2216697	Cable, RJ11 (25 feet long)
2255874	Cable, RJ11 Modular, 6-Conductor (7 feet long)
2249450	Global Modem Adapter Assembly for Europe
2276568	Cable, (940) MSM3-RS-232 to MSM1-J1

2.2 Mobile Satellite Modem Collector, Part 2385399

Table 2-3: Mobile Satellite Modem Collector Contents

Part Number	Item
2376563	Antenna Assembly
2376870	Satellite Modem, Power Supply and Battery assembled on mounting plate
2392246	Software, Satellite Modem Communication

2.3 Magmon3 Remote Alarm Collector, Part 5180301

Table 2-4: Magmon3 Remote Alarm Collector Contents

Part Number	Item/Cable Run	From	To
2266628	Magnet Monitor Remote Alarm Box		
2263160	Cable 915	MSM1A2-J1	FJ6
2263159	Cable 916	MSM1-J7	FJ1, FJ5, FJ6

2.4 Magmon3 LCC Cable Collector, Part 2350809-2

Table 2-5: Magmon3 LCC Cable Collector Contents

Part Number	Cable Run / Item	From	To	Comments
46-328000G979	823	MSM1-J10	MR2-A11-J24	
46-317359G975	824TR	PP1-J10	To FJ1	
46-317359G977	825TR	PP1-J48	To MSM1-A1-J8	
46-328000G978	826	FJ3	To MSM1-A1-J9	
46-328578P1	827	FJ3	FJ4 & MS5A5	
46-328000G974	828	MS1-A3-A1-J403	PP1-J10	
46-328000G976	829	MS1-FJ2	PP1-J48	
2222797	830	Magnet Room Interface		
46-328000G980	831	FJ1	MSM1-A1-J7	
46-328000G981	833	FJ4	MS5-A1-A6-JR	For Sumitomo Compressor
2271497	942	MSM1-J5	Network Wall Jack	
2333825	Lake Monitor; Water Flow / Temperature Sensor Assembly; Turbine Type			
46-265067P1	Screw lock connector to join cable connectors; (Quantity 10)			

2.5 Magmon3 Non-Zero Boil-Off Cable Collector, Part 2350809-3

Table 2-6: Magmon3 Non-Zero Boil-Off Cable Collector Contents

Part Number	Cable Run / Description	From	To	Comments
46-328000G979	823	MSM1-J10	MR2-A11-J24	
46-317359G975	824TR	PP1-J10	FJ1	
46-317359G977	825TR	PP1-J48	MSM1-A1-J8	
46-328000G978	826	FJ3	MSM1-A1-J9	
46-328578P1	827	FJ3	FJ4 & MS5A5	
46-328000G974	828	MS1-A3-A1-J403	PP1-J10	
46-328000G976	829	MS1-FJ2	PP1-J48	
2108726	830	Magnet Room Interface		
46-328000G980	831	FJ1	MSM1-A1-J7	
2333825	Water Flow / Temperature Sensor Assembly			
2299843-2	SUNX, Pressure Transducer			
2369032	RF Copper Shield, use on pressure transducer			Secure with 2 screws
46-260793P11	# 4-40 x 0.25 inch Round Head Brass Screw			Qty 2, use on RF Copper Shield
46-294167P98	Nipple, Pipe, Brass, 0.125 NPT Sch.40 x 1.50 Long			Adapts pressure transducer to plumbing
46-252294P1	0.25 inch Union Tee			Swagelok part B-400-3
2109440	0.25 inch port connector			Swagelok part B-401-PC
5180922	Union elbow; 0.25 inch Tube Fitting			Swagelok part B-400-9
5181130	0.125 in Male NPT x 0.25 in port connector adaptor;			Swagelok part B-4-TA-1-2
2204485	J24 Cable Magnet Monitor, System Cabinet – TPS			Use in System Cabinet with TPS
	Connector A (15 Pin sub D, MR2 A11 J24)			
	Connector B (SMA Plug, MR2 IPG J9)			
	Connector C (BNC Plug, MR2 PDU J5)			

**NOTICE**

The compressor interface cable must be ordered separately. This cable is not included in the cable collector. Part numbers are listed below.

Table 2-7: Compressor Interface Cables

Part Number	Use With
46-328000G981	Sumitomo compressor
2116875	Leybold RW4000 compressor
2404555	Leybold CoolPak compressor

2.6 Magmon3 HFO Cable Collector, Part 2265809-3

Table 2-8: Magmon3 HFO Cable Collector Contents

Part Number	Cable Run	From	To	Comments
2263200-41	823	MSM1-J10	MR2-A11-J24	
46-328000G975	824	FJ1	PP1-J10	
2263200-40	825	MSM1-J8	PP1-J48	
46-328000G978	826	MSM1-J9	FJ3	
46-328578P1	827 827b	FJ3 FJ3b	FJ4 & MS5A5 FJ4b & MS8A5	Qty 2 of Run 827, advise to mark the second one as Run 827b.
46-328000G976	829	PP1-J48	MS1-FJ2	
2279237	830	Magnet room interface		Used for monitoring 2 Coldheads.
2271499	832	Coldhead adaptor cable		
46-328000G981	833 833b	FJ4 FJ4b	MS5-A1-A6-JR MS8-A1-A6-JR	Qty 2 of Run 833 Same cable used for each compressor
2263158	913	PP1-J62	MS1-A1-A4-P302-3	
2262699	914	FJ5	PP1-J62	
2264859	919	PP1-J10	MS1-A1-A4-P302-1	
2276569	939	MDP-UPS OUT	MSM4	
2271497	942 OR	MSM1-J5 MSM1-J5	Network wall jack to OW1-A12-J7	If modem is used or if site network is not available.
2404562	1307	MSM1-J12	FJ3 and FJ7	For second compressor
2279530	n/a	Strain Relief - UPS Input Power Cable		Used on MDP for cable entry
2279530-2	n/a	Strain Relief - UPS Output Power Cable		Used on MDP for cable entry
2279530-3	n/a	Strain Relief - MUX Box Power Cable		Used on MDP for cable entry
2279530-4	n/a	Strain Relief - Global Modem/Magmon3 Power Cable		Used on MDP for cable entry
46-265067P1	n/a	Screw Lock Connector		(Qty 20) for joining cable connectors together
2333825	n/a	Water Flow / Temperature Meter		(Qty 2) One for each compressor

2.7 Magmon3 Cables Part of GEMSO 7.0T Cable Collector

Table 2-9: Magmon3 Cables Part of GEMSO 7.0T Cable Collector Contents

Part Number	Cable Run	From	To	Comments
46-271603G29	823	MSM1-J10	MR2-A11-J24	System Cabinet
46-328578P1	827 827b	FJ3 FJ3b	FJ4 & MS5A5 FJ4b & MS8A5	Qty 2 of Run 827, advise to mark the second one as Run 827b.
2222797	830	PP3-J9	MS1 A2-J2 MG3 TB1 MS1-A5-A2-J1	To RUO Pre-Amp To Reed Switch To Pressure Transducer
2263160	915	FJ6	MSM2-J1	Remote Alarm Box
2263159	916	MSM1-J7	FJ1 FJ5 FJ6	
2271497	942/1077	MSM1-J5	Site Network	Customer provided wall receptacle
2404555	1278 1278b	FJ4a (827a) FJ4b (827b)	MS5-A1-JR MS8-AI-JR	Qty 2 Run 1278, FE to mark the second one as Run 1278b
46-271600G37	1301	MSM1-J8	PP3-J9	
46-271600G37	1302	FJ7 (1307)	PP3-J10	
2404562	1307	MSM1-J12	FJ7 (1302) FJ3b (827b)	For SE Coldhead #2 For SE Compressor #2
2404554	1308	PP3-J10	MS1-A3-J4	To Service End Coldhead
2404553	1309	MSM1-J9	Lock Coil Timer FJ3a (827a)	For PE Compressor #1
46-271609G37	1310	Lock Coil Timer	PP3-J2	Mezzanine Room Pen Panel
2404561	1311	PP3-J2	Lock (1)	To Magnet Instrumentation Box
46-271601G37	1312	FJ5 (916)	PP3-J11	
2404558	1313	PP3-J11	LHe (1)	To Magnet Instrumentation Box
5183297	1410	MS1-A2-J1	MS1-A1-J6 MS1-A4-J2 MS1-A4-J3	Patient End Coldhead Vacuum port; left side; Service End Vacuum port; right side; Patient End

2.8 Magmon3 Cable Collector - Part 5193795

Table 2-10: Magmon3 MR750/MR450/MR450w Cable Collector Contents

Part Number	Cable Run / Item	From	To	Comments
5357236	M3023	SPW-J117	MAG-FJ2	Secondary Penetration Wall (SPW) to Magnet
5351557	E3023	SPW-J117	MON-J8	SPW to Monitor
5351627	E3022	MON-J7	SPW-J116	To SPW Instrumentation Box
5352885	E3024	MON-J9	MON-FJ4 and MON-FJ3	Flow Monitoring
5352519	E3026	HEC-FL-Temp	MON-FJ3	To Compressor Interface
5351365	E3020	PEN-J13	MON-J10	PEN Cabinet to Monitor
5351806	E3025	CRY	MON-FJ4	To Flow Monitor-Sumitomo
5351746	M3022	SPW-J116	MAG-P403	SPW to Magnet
5351608	E3034	MON-J5	Network	Monitor to Network
5115594-3	N/A	Power Cord IEC 60320 C13	NEMA 5-15P	
5352964	830	Magnet Interfaces	Cable Interface, Run 829	To Magnet Room Interface

Chapter 3 Installation

1 Magmon3 Installation

1.1 Introduction

1.1.1 Features Overview

The Magmon3 consists of the front panel display, user keypad hardware, and software used to:

- Measure helium vessel level
- Measure helium vessel pressure
- Provide helium vessel pressure control on magnets that have recondenser systems, i.e., LCC and HFO magnets
- Measure coldhead and magnet shield/recondensor temperatures
- Measure coldhead cycling (for HFO systems)
- Monitor two coldheads and two compressors
- Measure water flow and temperature for the water-cooled compressor(s)
- Monitor status of compressors
- Auto restart the compressor(s) after interruption in power or cooling
- Detect the loss of the magnet field

1.1.2 Magnet Compatibility

The Magmon3 is designed to interface with the following GE-manufactured magnets:

- S2 through S5 - 1.5T
- MR Max - 0.5T (if installed on Signa System)
- Compact Magnet - 0.5T (if installed on Signa System)
- SX - 1.0T (including Mobile)
- CX - 1.0T / 1.5T (including Mobile)
- LCC - 1.0T / 1.5T (including Mobile)
- LCC - RB 1.5T
- DVw - 1.5T
- LCC300 - 3.0T
- HFO - 0.7T

- GE Oxford - 7.0T

1.1.3 Shield Cooler / Cryocooler Compressor Compatibility

The Magmon3 is designed to interface with the following compressors:

- Leybold Models: RW4000, RW4200, RW6000, RW6200, CP4000, CP4200 and CP6000
- Sumitomo Models: CSA-71A, CSW-71C, CSW-71D, CNA-61C & D, F-50, and HC10

NOTE: A water flow / water temperature sensor is externally installed in the coolant supply line except for air cooled compressors.

1.1.4 Magmon3 System and Parts Structure

The Magmon3 device, sensors and cables are shipped with Signa product line systems. Refer to [Chapter 2, Magmon3 Collector Structures](#).

1.1.5 Optional Parts

The Magmon3 Kit structures provide most of the devices and cables needed during installation. However, one or more of the following items may need to be obtained, depending on system requirements and needs.

Additional parts may be needed for non-zero boil-off magnet. All are stocked and can be ordered through normal GPRS channels.

Table 3-1: Additional Parts for Non-Zero Boil-Off Magnet

Part Number	Description	Where Used
46-328000G981	(833) FJ4 TO MS5-A1-A6-JR (used for Sumitomo Compressor)	Sumitomo CSW-71A, CSW-71C, CNA-61C & D, F-50, and HC-10 Compressors
2116875	(833B) FJ4 TO MS5-A1-A6-JR, (used on Leybold RW4000 Compressors)	Leybold RW 4000, RW4200, RW6000 and RW6200 Compressors
2404555	(1278) FJ4 to MS5-A1-JR (used on Leybold CoolPak Compressors)	Leybold CoolPak (CP) 4000, 4200, 6200, and 6200D Compressors

Table 3-2: Additional Parts Not Related to Shield Cooler System

Part Number	Description	Where Used
2394952	Magnet Monitor 3 with power cord; Laptop crossover network cable; Installation/Operator Manual	All GE Magnets, S2 forward
46-318042P1	(627) Lakeshore Meter Cable 11 ft. long; Coldhead Sleeve to Rear Pedestal for measuring shield temperatures	All GE Magnets, S2 through CX
2204485	J24 Cable Magnet Monitor, System Cabinet - TPS, Connector A (15-Pin Sub D, MR2 A11 J24); Connector B (SMA Plug, MR2 IPG J9); Connector C (BNC Plug, MR2 PDU J5)	LX Systems Cabinet with TPS Chassis
46-296816P1	Adapter, BNC Tee, 50-ohm Amphenol #31-200 - All Center Conductors are Silver Plated (Rev Tk, 11/91). Has Rexolite Insulation. Used with System Cabinet cable 2204485 for MR2-PDU-J5.	LX Systems Cabinet with TPS Chassis

46-265067P1	I/F Panel Cable Screw Lock Connector	All System Cabinets
2322581	For System Cabinets with an MGD Chassis: Order interface cable P/N 2322581.	Excite Systems Cabinet with MGD Chassis
2245794-2	MagMon3 Global Modem Kit	All monitored sites as needed
2271497	(942) Ethernet Cable	All Magnets
46-271920P3	25-Pin Penetration Panel Filter	All Magnets - Use when 25-Pin filter is not available on Pen Panel.
46-221754P3	Penetration Panel Filter Gasket	All Magnets - Use when 25-Pin filter is not available on Pen Panel.

Additional cables for sites where the Penetration Panel is more than 60 ft. away from the Magmon:

- 25-Pin Male-Female, 10 ft. Cable (46-271601G19)
- 25-Pin Male-Female, 20 ft. Cable (46-271601G21)
- 25-Pin Male-Female, 30 ft. Cable (46-271601G23)
- 25-Pin Male-Female, 40 ft. Cable (46-271601G25)
- 25-Pin Male-Female, 50 ft. Cable (46-271601G27)

Modem-Based Systems

1. Multiplexer with four station lines:
 - 115 VAC Power Supply (46-328475P1)
 - 220 VAC, 50 Hz Power Supply (46-328475P3)
2. Multiplexer with eight station lines:
 - 115 VAC Power Supply (46-328475P2)
 - 220 VAC, 50 Hz Power Supply (46-328475P4)

NOTE: On OpenSpeed, TwinSpeed and Signa 7.0T, there are dedicated 115 VAC outlets inside the Main Disconnect Panel (MDP) for powering the Magmon, Modem and Multiplexer Box.

1.1.6 Installation and Setup Sequence

This is the recommended sequence of steps for installing a new Magmon3.

1. Perform inventory of all hardware prior to installation. Refer to [Chapter 2, Collector Structures](#).
2. Install the Magmon3 unit. Refer to [Section 1.3](#) and [Chapter 1, Component Specifications](#).
3. Install the Modem (not used on all systems). Refer to [Section 1.3.2](#) and to [Chapter 1, Component Specifications](#).

4. Install the Remote Alarm Box (not used on all systems). Refer to [Section 1.3.14](#) and to [Chapter 1, Component Specifications](#).
5. **(For 7.0T systems)** Physically install the Lock Coil Timer. Refer to Direction 2357818, *GE HCO 7.0T/900 High Field Magnet and Cryogen Subsystems*.
6. Install the water flow / temperature meter(s). Refer to [Section 1.3.3](#) and to [Chapter 1, Component Specifications](#).
7. Install the pressure transducer on the magnet (installed base upgrades only). Refer to [Section 1.3.4](#) and to [Chapter 1, Component Specifications](#).
8. Install and connect Magmon3 system cables. Refer to [Chapter 4, Interconnect Diagrams](#).
9. Apply AC power to the Magmon3 and Modem (if used). Refer to [Chapter 4, Interconnect Diagrams](#).

NOTE: On OpenSpeed, TwinSpeed and Signa 7.0T, there are dedicated 115 VAC outlets inside the Main Disconnect Panel (MDP) for powering the Magmon, Modem and Multiplexer Box. For all other systems, the Magmon3 will plug into customer-provided 7 x 24 wall outlets.
10. Become familiar with the front panel keypad and user display. Refer to [Chapter 5, User Interface Panel](#).
11. Connect a laptop to the Magmon3. Refer to [Chapter 6, Laptop Connectivity](#).
12. Set the date and time on the Magmon3. Refer to [Chapter 9, Time and Date](#).
13. Select the correct magnet type. Refer to [Chapter 9, Magnet Type](#).
14. Establish the correct network settings on the *Network* page. Refer to [Chapter 9, Network](#).
15. Click [Logout] on the top left side of the web page to activate the new settings.
16. Wait at least ten minutes, then check for alarms on the front panel as indicated by the alarm LED and the [Alarms] button. Refer to [Chapter 5, User Interface Panel](#).

1.2 Tools and Equipment

1.2.1 Required Personal Protective Equipment (PPE) and Safety Items

Table 3-3: Required PPE and Safety Items

Item	Description
1	Safety Glasses
2	Safety Shoes
3	Pair of Gloves
4	Restricted Access Signage

1.2.2 Required Tools and Equipment

The following tools are required to install the Magmon3:

- Nonmagnetic Adjustable Wrench
- Nonmagnetic 9/16 inch Combination Wrench
- Nonmagnetic Flat-Blade Screwdriver
- Ruler
- Utility Knife
- Nut Drivers, 3/16 and 1/4 inch sizes
- Helium Resistance Box (46-265286G1)
- Digital Volt Meter (DVM)
- No. 2 Phillips Screwdriver
- Static Wrist Strap
- Shallow Container (to catch water after disconnecting the hose to the Shield Cooler Compressor)
- Tools and hardware needed to mount Magmon to wall such as: power drill, drill bits, wall anchor system, etc.
- Laptop computer (minimum requirement is Windows 2000 with Internet Explorer or Netscape Navigator)
- Crossover Network Cable (supplied with every Magmon3 device)
- Six M4 x 8 to 20 mm Phillips Pan-Head Screws for mounting to Heat Exchange Cabinet (HEC), not included

1.3 Hardware Installation

1.3.1 Magmon3 Wall Installation

For MR750, MR450, and MR450w systems, the Magmon3 should be attached to the HEC. See [Illustration 3-1](#).

Illustration 3-1: Magmon3 Mounting on HEC



The Magmon3 unit is designed for wall mounting. The location of the unit within the Equipment Room on new system installations should be shown on the customer's architectural drawings. For Signa OpenSpeed, Signa TwinSpeed and Signa 7.0T systems, the Magmon3 device is mounted

next to the Main Disconnect Panel (MDP) because there are 120 VAC receptacles inside the MDP designated for the Magnet Monitor and a modem.

- For systems where the GE MDP is not included, the customer is required to provide a dedicated circuit that will supply the Magmon with 7 x 24 AC power.
- For installed base systems receiving a Magmon3 for the first time, make sure the location complies with the equipment specifications for Gauss levels and service clearance. Refer to [Chapter 1, Component Specifications](#).
- For installed base systems that already have a Magmon2 (SHeM manufactured by Granite) and are upgrading to a Magmon3, mark new hole locations on the wall because the mounting hole patterns are different. Refer to [Chapter 1, Component Specifications](#) for mounting hole dimensions.

Table 3-4: Field-Supplied Items

Item	Description	Quantity
1	Electric Hand Drill	1
2	Assorted Drill Bits (size is dependent on wall anchor system)	As required
3	Wall Anchor Hardware (to mount the Magmon3 device)	4
4	Hand Tools, Screws, Socket Drive (size dependent on wall anchor system)	As required
5	Tape Measure or Ruler (for laying out the hole locations)	1

Table 3-5: Personal Protective Equipment

Item	Description	Quantity
1	Safety Glasses	1 pair
2	Protective Gloves	1 pair

Anchor the Magmon to the wall using all applicable codes for the site. For customers without the GE MDP, make sure a nonswitched 120/240 VAC power is within 6 feet of the mounting location of the Magmon.

- The Magmon3 automatically adjusts to line voltage and line frequency, so no further adjustments are required.
- Do **not** connect the AC power cord to the Magmon3 until all cabling is completed. (The Magmon3 does not have an ON-OFF switch for power control.)
- If preferred, the Magmon3 may be plugged into a 220 VAC outlet. Obtain a power cord with the appropriate connector.

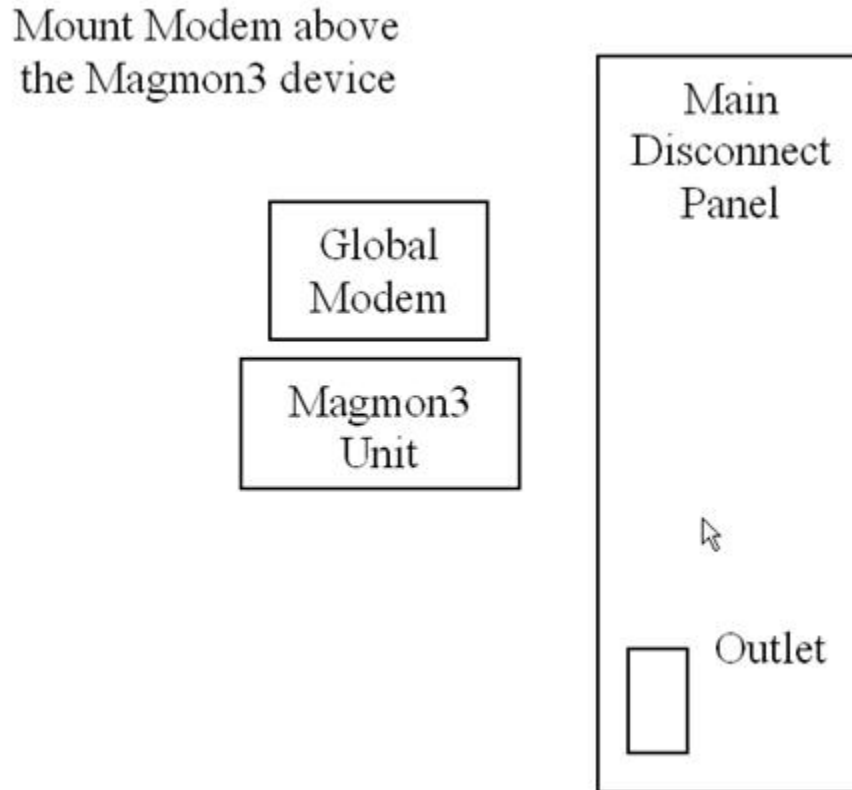
1.3.2 Global Modem Installation (Optional)

If broadband connectivity is not available, a Global Modem is used for connectivity to the back office. The Global Modem will source its power from the same receptacle as the Magmon3.

For Signa OpenSpeed, Signa TwinSpeed and Signa 7.0T systems, the Global Modem device is mounted above the Magmon3 and next to the Main Disconnect Panel (MDP) because there are

120 VAC receptacles inside the MDP designated for the Modem and Magnet Monitor. See [Illustration 3-2](#).

Illustration 3-2: Magmon3 / Modem Powered from MDP



For systems where the GE-designed MDP is not installed, the customer is required to provide a dedicated circuit that will supply the Magnet Monitor and Modem with 7 x 24 AC power.

1. Use supplied Velcro to secure the modem to the wall above the Magnet Monitor.
2. Use the proper power adapters for the power connection from the wall to the power supply. (The modem power supply will work for any voltage and frequency.)
3. Connect Cable Run 940, provided with the Global Modem Kit, between the Magnet Monitor J1 and Modem Serial Connector.

1.3.3 Water Flow / Temperature Meter Installation

The turbine style water flow / temperature meter is designed to be installed in the supply waterline for all compressors except air cooled compressors.

Table 3-6: Field-Supplied Items

Item	Description	Quantity
1	Five-gallon Pale (empty)	1
2	Rags or Towels	3 to 5
3	Electric Hand Drill	1

4	Assorted Drill Bits	As required
5	Mounting Clamps (for 1/2 brass pipe)	2 per flow meter
6	Wall Anchor Hardware (to support the flow meter)	2 per flow meter
7	Hand Tools, Screws, Socket Drive (size dependent on wall anchor system)	As required

Table 3-7: Personal Protective Equipment

Item	Description	Quantity
1	Safety Glasses	1 pair
2	Rubber Gloves	1 pair

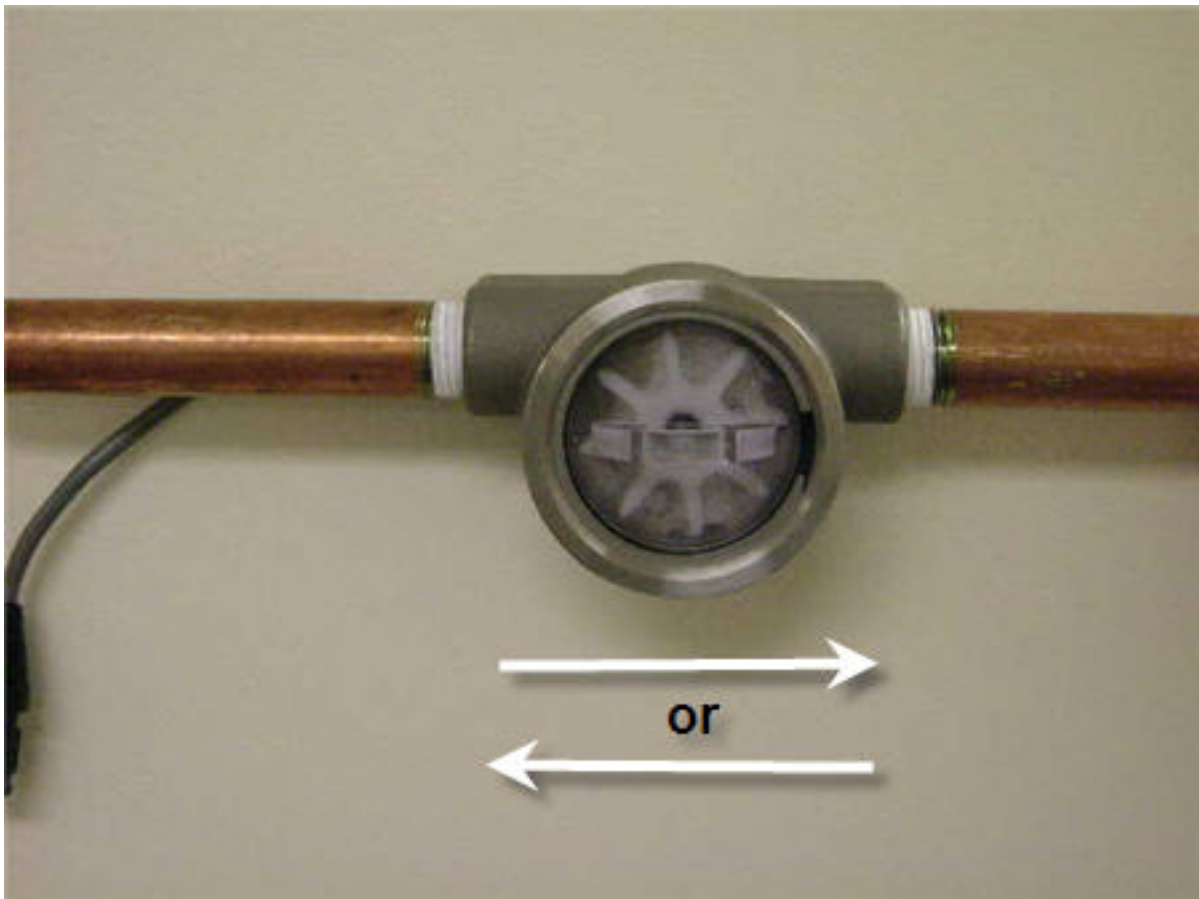
**NOTICE**

The flow meter must be mounted horizontally with the inlet and outlet piping located at the top as shown in [Illustration 3-3](#). This orientation allows any trapped air in the lines to pass through the meter cavity.

Do NOT remove the 5-inch length of copper pipe attached to inlet and outlet ports of the meter. This straight length of pipe is required to minimize turbulent flow through the meter, which impacts the performance of the meter.

The flow meter is bi-directional; it does not matter which end is used for the inlet and outlet.

Illustration 3-3: Flow Meter Orientation



1. Mount the flow meter to any vertical surface. Do **not** lay the meter flat on the floor, which will result in excessive impeller wear and degrade the performance of the unit over time.
2. Shut off the water flow to the Coldhead Compressor, and make sure to shut off both the supply and return lines to minimize water spillage.
3. Place a container under the inlet water line hose to catch any water that may run off, and cut the hose far enough back from the compressor to allow connections to be made to the hose barb connectors on the water flow meter.
4. Connect the cut hose to the inlet and outlet sides of the water flow meter and secure with a hose clamp.
5. Turn on the water supply and return line valves; check all fittings for leaks, and repair leaks as necessary.

**NOTICE**

If water from the compressor spills, clean it up following hospital procedures as the water may contain chemicals, such as glycol.

1.3.4 Pressure Transducer Installation

The pressure transducer ships already installed on all forward production magnets. This procedure is intended for installed base magnets where a pressure transducer is not installed.

1. Locate the pressure transducer and assorted plumbing fittings in the Magmon3 Upgrade Kit.
2. Attach the pressure transducer to the vent plumbing on the magnet just after the mechanical pressure gauge, using the appropriate plumbing adaptors.
3. Leak test the plumbing fitting, and correct any leaks as needed.

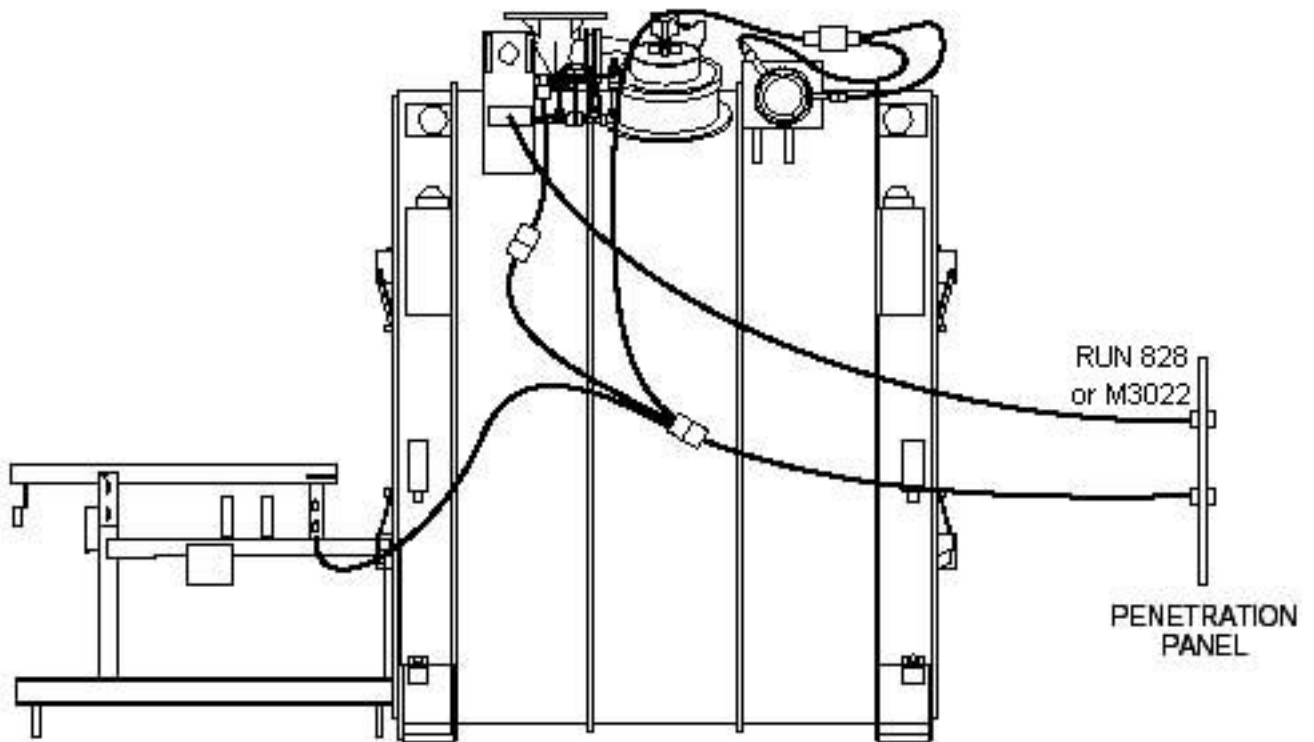
1.3.5 Connecting Magnet Room Cables (All Systems)

The Pressure Transducer, Magnet and Coldhead Temperature Sensors, and Reed Switch generate signals incorporated into a cable that enters the Magnet Room through the Penetration Panel. These signals aid in assessing boil off, magnet quench and proper maintenance of vessel pressures before, during, and after a helium fill.

Most of the hardware installation can be performed while the customer is using the system. The Magnet Monitor can be shut down and worked on while the Signa System is running.

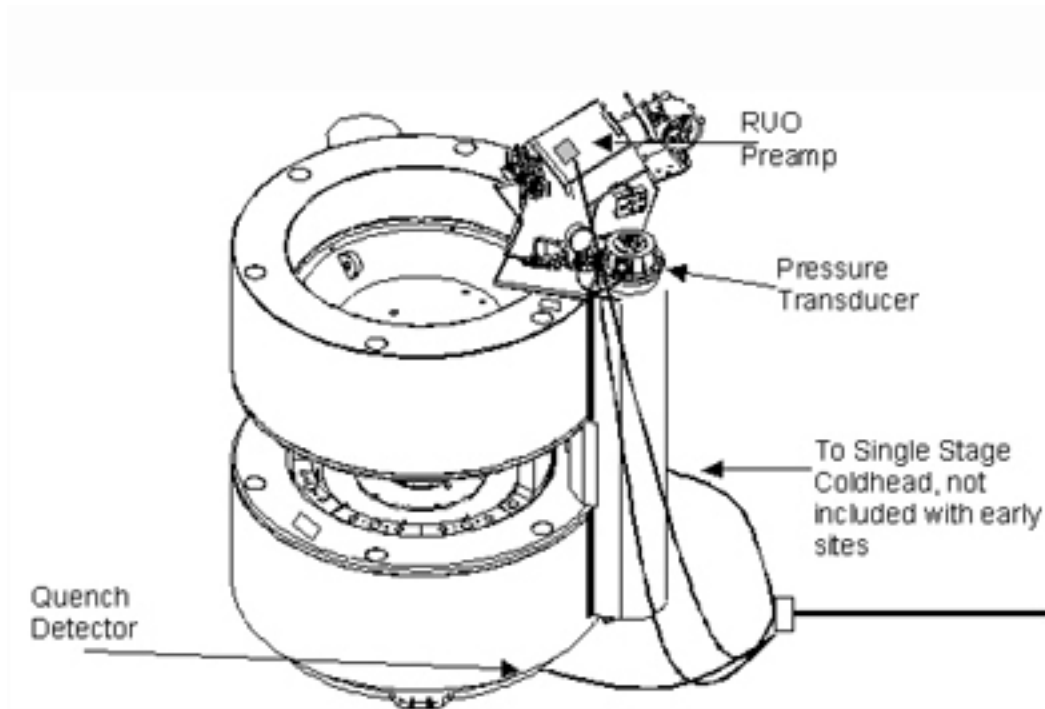
- The Magnet Room is needed for about one hour to install cables and sensors. Remove the twist locks around the cables before connecting cables in the Magnet Room.
 - The three to four hours of installation time in the Equipment Room do not require the system to be down.
1. Refer to [Chapter 4, Interconnect Maps](#), and use the appropriate interconnect map for the magnet system being worked on.

Illustration 3-4: LCC and DVw Magnet Room Cabling



2. Refer to [Illustration 3-4](#) for LCC cable routing. Refer to [Illustration 3-5](#) for typical HFO cable routing.

Illustration 3-5: HFO Sensor Cabling



HFO Magnet Room Cabling

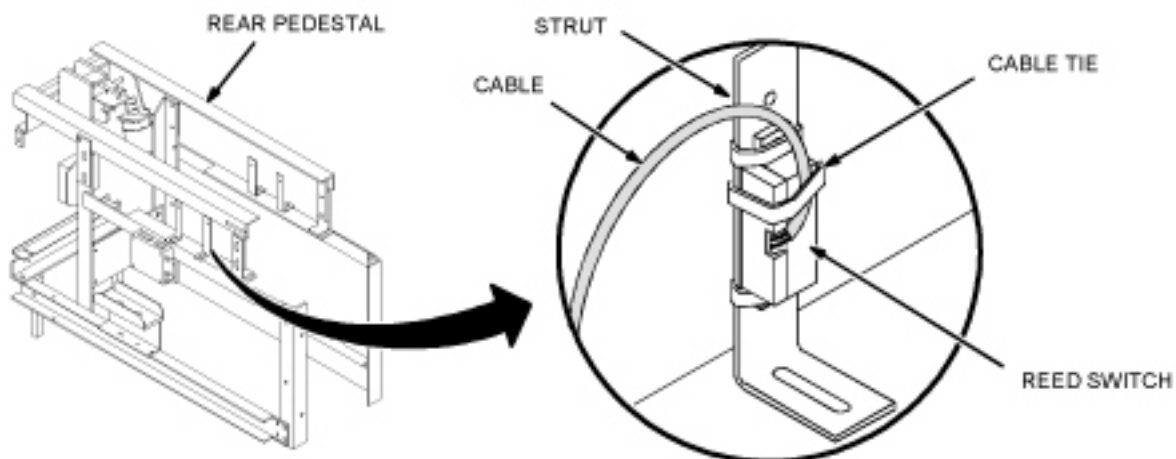
NOTE: Do **not** coil the cables under the rear pedestal. Serpentine any excess cable in the cable trough so it will not interfere with scanning.

1.3.6 Connecting Reed Switch (All Magnets Except HFO)

NOTE: The Magmon3 does not use this Reed switch. Installation is optional. The magnet monitor does not initiate any alarms with the change in state of the reed switch. Magnet Monitor kits manufactured for the DV platform systems after March 2010 do not include the reed switch.

1. Position the Reed Switch where it detects the magnetic field. The contacts are Open when the Reed Switch detects a magnetic field. (Usually, the switch can be placed vertically on one of the support struts on the rear pedestal. See [Illustration 3-6](#).)
2. Wrap two Cable Ties (46-208758P3) around the top and bottom of the Reed Switch, and the support strut to hold the switch in place. See [Illustration 3-6](#).
3. Use an additional cable tie to secure the cable to the strut to prevent the wires from being pulled out of the switch, and move the ends of the cable ties to the back of the strut.

Illustration 3-6: Reed Switch Placement



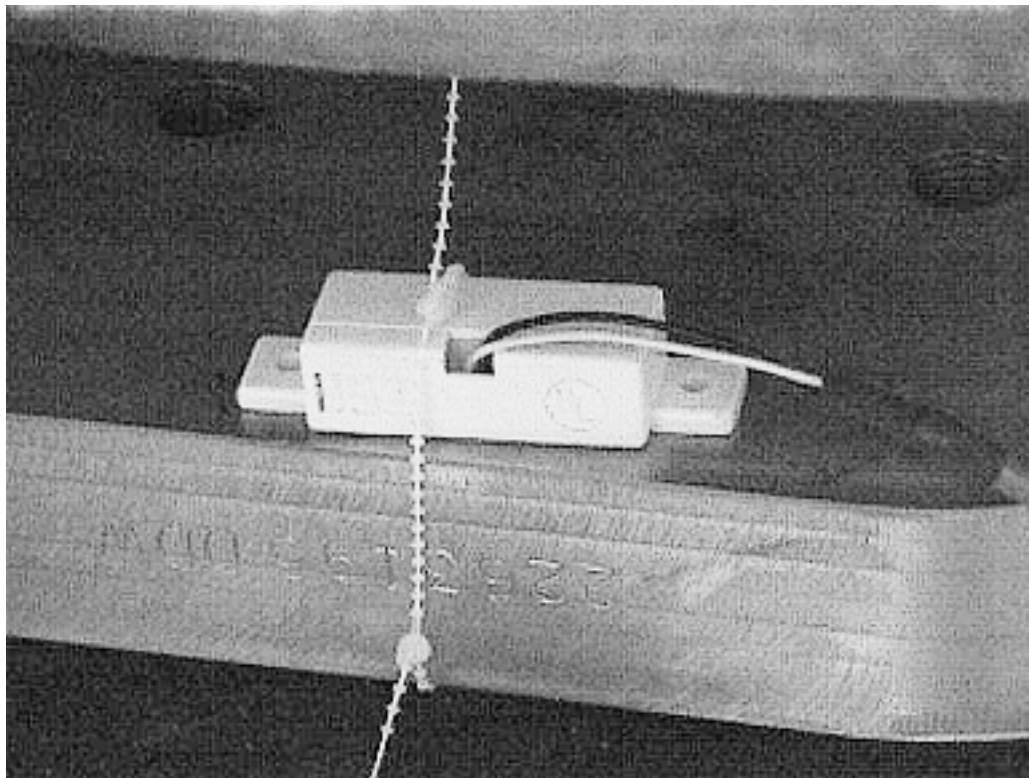
1.3.7 Connecting Reed Switch (HFO Systems Only)

NOTE: The Magmon3 does not use this Reed switch. Installation is optional.

NOTE: The Reed Switch (2108790) is pre-assembled to Run 830, the Magnet Room Interface Cable (2222797).

1. Position the Reed Switch on the bottom plate of the magnet, and secure the Reed Switch to the magnet using cable ties through an open bolt hole as shown in [Illustration 3-7](#). Make sure the orientation of the switch is the same as shown in [Illustration 3-7](#) to properly detect the magnetic field.

Illustration 3-7: Reed Switch Placement HFO

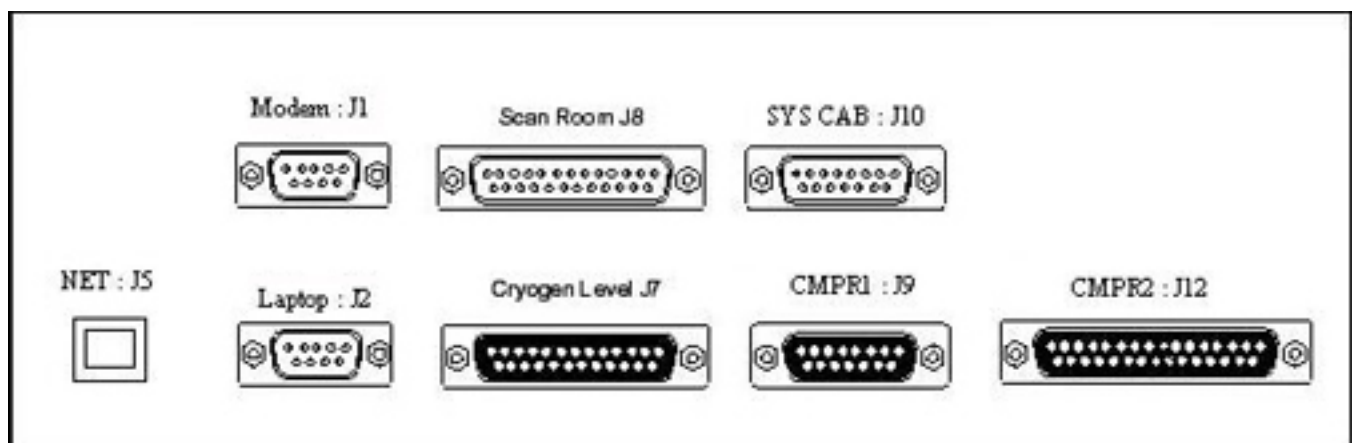


2. Use an additional cable tie to secure the cable to the magnet and prevent the wires from being pulled out of the Reed Switch.

1.3.8 Connecting Equipment Room Cables (All Systems)

1. Refer to [Chapter 4, Interconnect Maps](#), and use the appropriate interconnect map for the magnet system being working on. See [Illustration 3-8](#).

Illustration 3-8: Magmon3 Interface Panel Connector Locations



2. Install and route cable from the Magmon3 J7 to the Pen Panel using the Run # identified on the interconnect map.

3. Install and route cable from the Magmon3 J8 to the Pen Panel using the Run # identified on the interconnect map.
4. Install and route cable from Magmon3 J9 to the Coldhead Compressor using the Run # identified on the interconnect map.
NOTE: For nonzero boil-off magnets, the Compressor Interface Cable must be ordered separately. See [Chapter 2, Non-Zero Boil-Off Cable Connector](#) for proper part numbers.
5. **(For HFO only)** Install and route cable from Magmon3 J12 to the Second Coldhead Compressor using the Run # identified on the interconnect map.
6. Install and route cable from Magmon3 J10 to the System Cabinet using Run 823.
7. Connect the power cord between Magmon3 and the customer-supplied 24 x 7 power source.

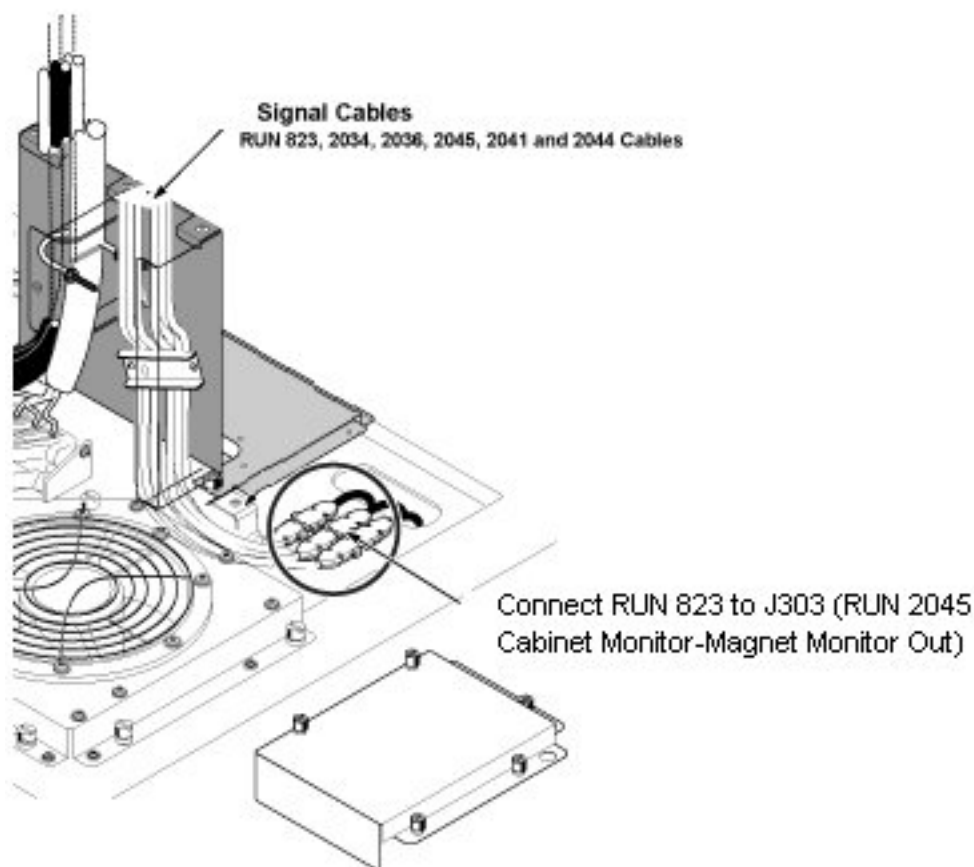
1.3.9 Connecting Magmon to System Cabinet (Signa HDe 1.5T)

Refer to Run 823 shown in [Illustration 3-9](#).

1. Connect the MSM1-A1-J10 end to the Magnet Monitor.
2. Route the cable to the top of the System Cabinet.
3. Connect the MR2-A11-J24 end of Run 823 to J303 (Run 2045 Cabinet Monitor, Magnet Monitor-Out Cable).

NOTE: For an HDe System, an unblank signal line is included in Run 2045 Cabinet Monitor, Magnet Monitor-Out Cable.

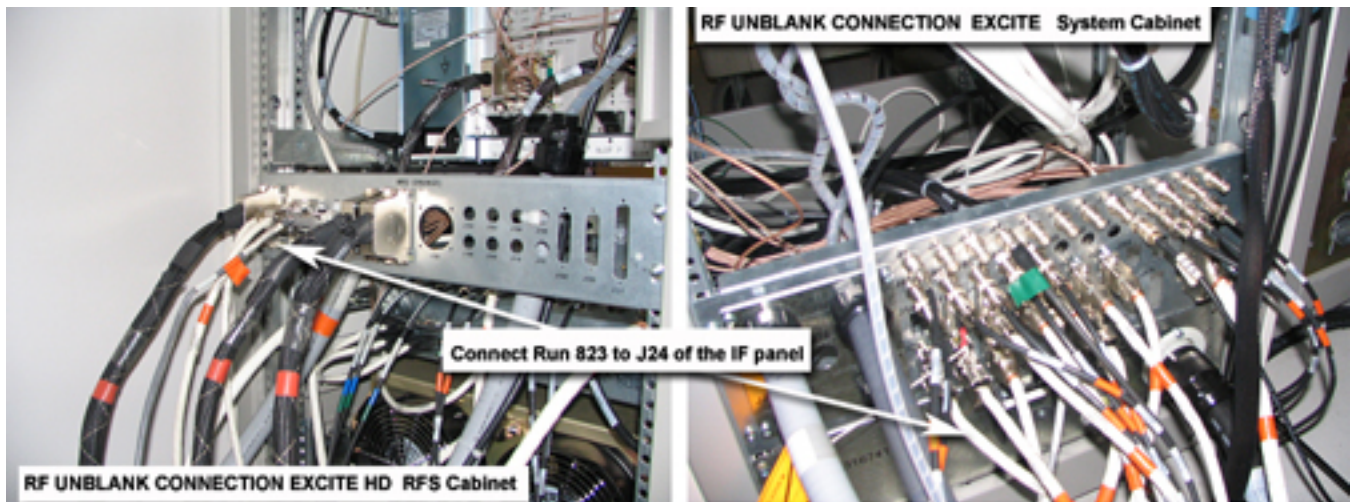
Illustration 3-9: Connect Run 823 (View from Top of System Cabinet)



1.3.10 Connecting Magmon to System Cabinet (LX and Excite-Based Systems)

1. Refer to Run 823 shown in [Illustration 3-10](#):
 - a. Connect the MSM1-A1-J10 end to the Magnet Monitor.
 - b. Route the cable to the interface panel at the lower rear of the System Cabinet / RF System Cabinet.
 - c. **(For Pre-Excite Systems)** It may be necessary to install a pair of Connector Joiners (46-265067P1) in an empty 15-Pin opening (J24 if available) on the interface panel.
 - d. Connect the MR2-A11-J24 end of Run 823 to the above.

Illustration 3-10: Connect Run 823



2. For Run X (no number) internal to an LX System Cabinet:

NOTE: All Excite System Cabinets have the internal cable pre-installed prior to shipping to the customer.

- a. Connect the MR2-A11-J24 end to the Interface Panel Connectors installed above.
- b. Carefully thread the MR2-IPG-J9 end through to the front of the System Cabinet, and connect to J9 (RF UNBLANK1) of the IPG Board on the front of the cabinet. See [Illustration 3-11](#).
- c. At MR2-PDU-J5 in the System Cabinet, remove the existing cable and replace it with a BNC Tee. Connect the existing cable and the MR2-PDU-J5 end of Run X to the BNC Tee. See [Illustration 3-12](#).
- d. Install the Jumper Plug (2239626) on the back of the Tyme Board in position J5.

NOTE: This jumper plug eliminates the *low helium* message displayed on the console. Some of the early sites may not have the jumper plug. Pins 4 and 6 of Connector J5 (Cryo-mon) can be shorted on the Tyme Board to eliminate the message.

Illustration 3-11: RF Unblank Connection

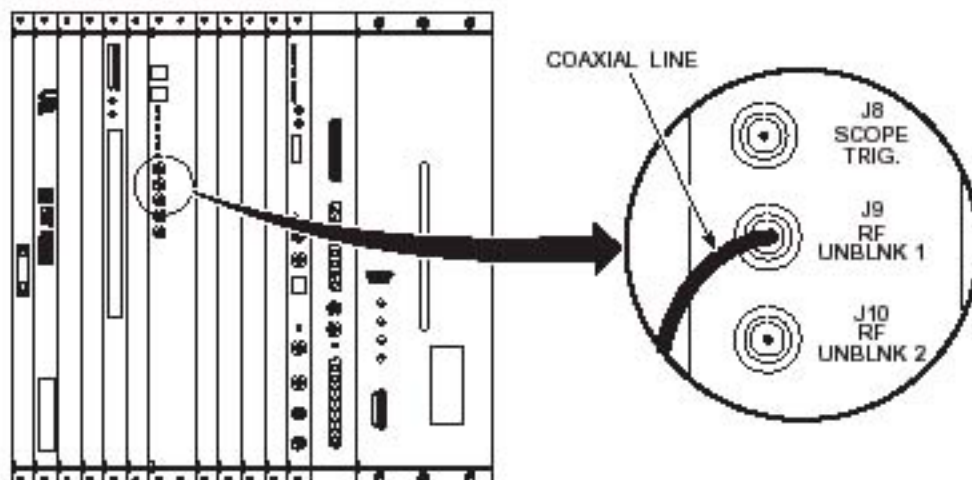
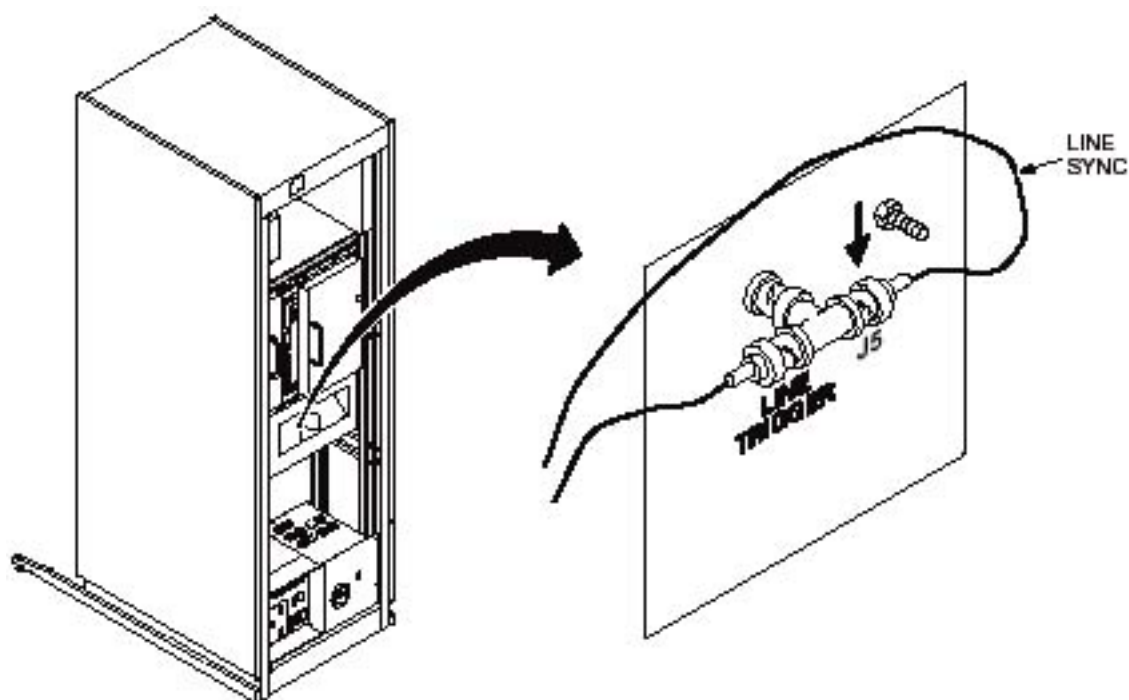


Illustration 3-12: Line Sync Connection



1.3.11 Connecting Magmon to System Cabinet (MR750, MR450, and MR450w Systems)

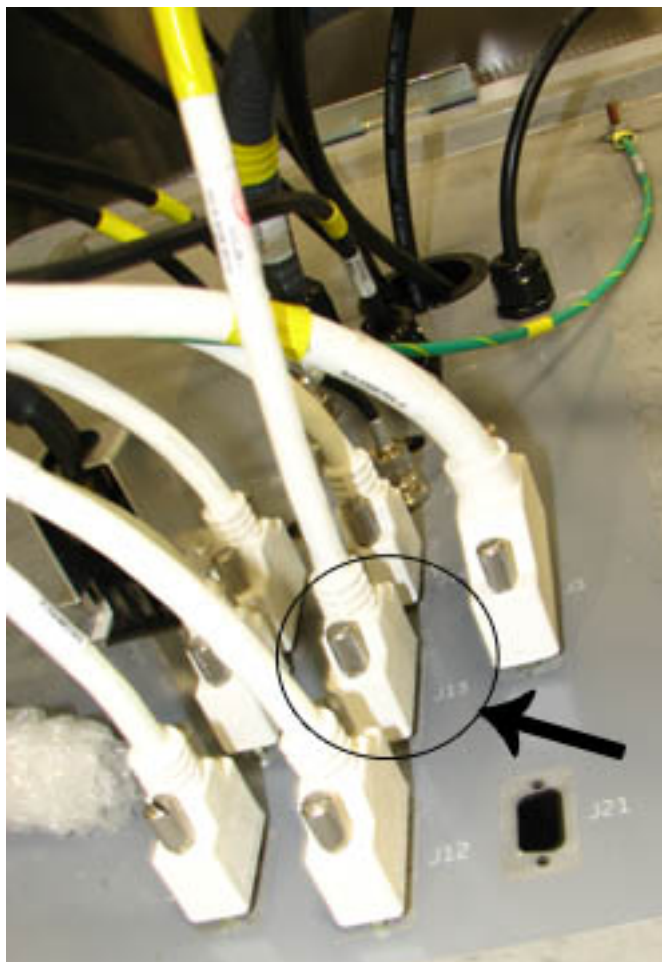
Connect Run E3020 as described in the following steps:

1. Connect the MON-J10 end of Run E3020 to the Magnet Monitor (see [Illustration 3-13](#)).
2. Route the cable to the interface panel at the top of the System Cabinet / RF System Cabinet.
3. Connect the PEN-J13 end of Run E3020 to the interface panel (see [Illustration 3-14](#)).

Illustration 3-13: Connect MON-J10 to the Magnet Monitor



Illustration 3-14: Connect PEN-J13 to the Interface Panel



1.3.12 Installing MUX Box (Optional for Dialup Only)

1. Find a suitable location for the MUX Box near the Magnet Monitor.
2. Connect the power cord for the MUX Box to the MDP into the AC outlet labeled **UPS power**.
3. Connect the customer phone line input (through the UPS) to the MUX Box port labeled **phone line in**, and connect the phone line between the Global Modem and the MUX Port 1.



NOTICE

Make sure Port 1 of the MUX Box is used for the Magnet Monitor. The MUX Box is NOT connected to the UPS. If the site were to lose power, the MUX Box will default to Port 1, which will allow the Magnet Monitor to dial out if the MUX Box loses power.

1.3.13 Installing Magmon UPS on HFO (Optional for Twin and 3.0T)

1. Remove the AC jumper between J1 and J2 in the Main Disconnect Panel (MDP). Locate the UPS In and UPS Out labels. Place the **UPS In** label over Outlet J2, and the **UPS Out** label over Outlet J1.
2. Position the UPS near the MDP and the Magmon; connect the power cable, provided with the UPS, to the MDP AC outlet labeled **UPS Out**. (The cable must be routed through the bottom of the MDP cabinet. See [Illustration 3-15](#).)

Illustration 3-15: MDP Connections



3. Find Cable Run 939 provided with the Magmon Cable Kit, and connect it from the UPS output to the MDP AC outlet labeled **UPS Input**. (The AC outlet in the MDP is a male connector. The cable must be routed through the bottom of the MDP cabinet.)

4. Connect the Magmon AC power cable to the MDP AC outlet labeled **Magnet Monitor Output**. (This cable must be routed through the bottom of the MDP cabinet.)
5. Connect the Customer Phone Line to the UPS **phone in** connection; run the customer supplied phone line from the UPS **phone out** connection to the global modem (only for dial-up).
6. Connect the provided strain reliefs for the power cables routed through the MDP panel (included in the Magmon Cable Kit).

1.3.14 Remote Alarm Box Installation

The Remote Alarm Box (optional for all other magnet types) is shipped with HFO 0.7T and the 7.0T Magnet Monitor Kits. Refer to [Chapter 1, Magmon3 Component Specifications](#) for hardware specifications.

1. Refer to [Chapter 4, Interconnect Maps](#), and use the appropriate interconnect map for the magnet system being worked on.
2. Choose an appropriate place on or near the Operator Workspace table for locating the Remote Alarm Box.
3. Mount the Remote Alarm Box in this location.
4. Route Cable Run 915 from the Magmon3 to the Remote Alarm Box.
5. Install the Adapter Cable Run 916 to Magmon3 J7.
6. Connect the other ends of the Run 916 as defined on the interconnect map.

1.4 Power Up and Helium Level Verification

1.4.1 Power Up

After verifying that all cabling and connections to the Magmon3, Modem (if phone connection is utilized), MUX Box (if used), Broadband (if network connection is used), and the Magnet are correct, power up the Magmon3 by plugging the power cable into Magmon3 J2.

NOTE: If used, the UPS does **not** require a serial connection to the Magmon3. The Magmon3 monitors magnet parameters to determine if power is lost.

1.4.2 Helium Level Verification

No calibration is required, but the helium reading verification check should be performed every year. Verify the Magmon3 helium level reading is tracking properly by performing the following tasks:

1. Obtain the Helium Meter Calibration Box, P/N 46-306864G1.
2. Observe the helium readings for the vessel.
3. Disconnect the helium level cable, located on the Magnet Interconnect Box, from the magnet.
4. Connect the Helium Meter Calibration Box to the Cable going to P302-Upper, and adjust the resistance on the Helium Meter Calibration Box to the values listed in [Table 3-8](#).
 - a. Press the [Sample] button on the Magmon.

- b. Verify that the helium reading is within the specification listed in [Table 3-8](#). For noncompliant devices, replace the unit.

Table 3-8: Resistance/Helium

Resistance	Value Nominal	Value Specification
0 ohms	100%	98 to 100%
230 ohms	66.4%	63.4 to 69.4%
460 ohms	32.7%	29.7 to 35.7%
686 ohms	0.0%	0 to 3%

Chapter 4 Interconnect Maps

1 Interconnect Maps

1.1 LCC 1.5T/3.0T - Fixed/Transportable/Relocatable

Illustration 4-1: LCC 1.5T/3.0T - Fixed/Transportable/Relocatable

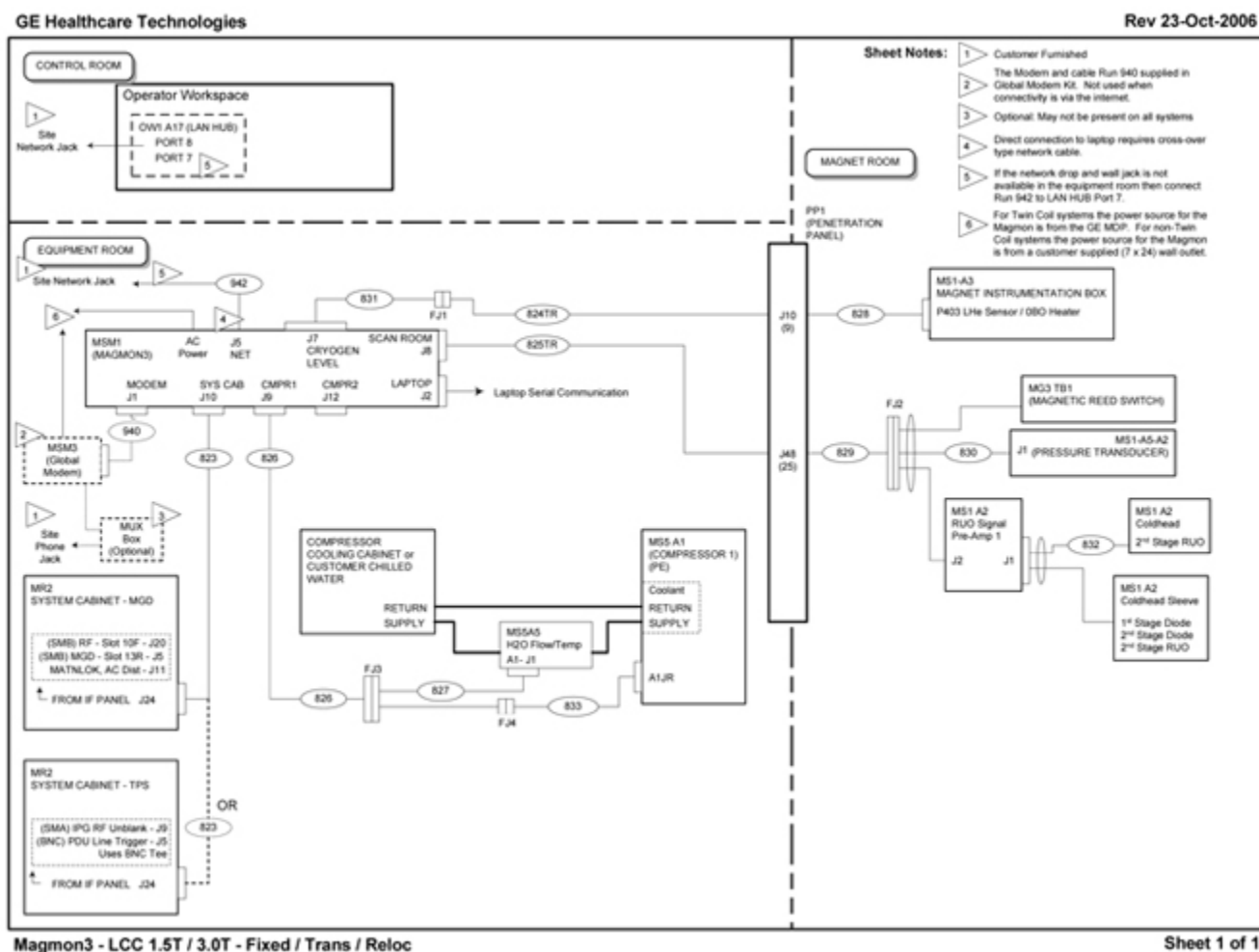
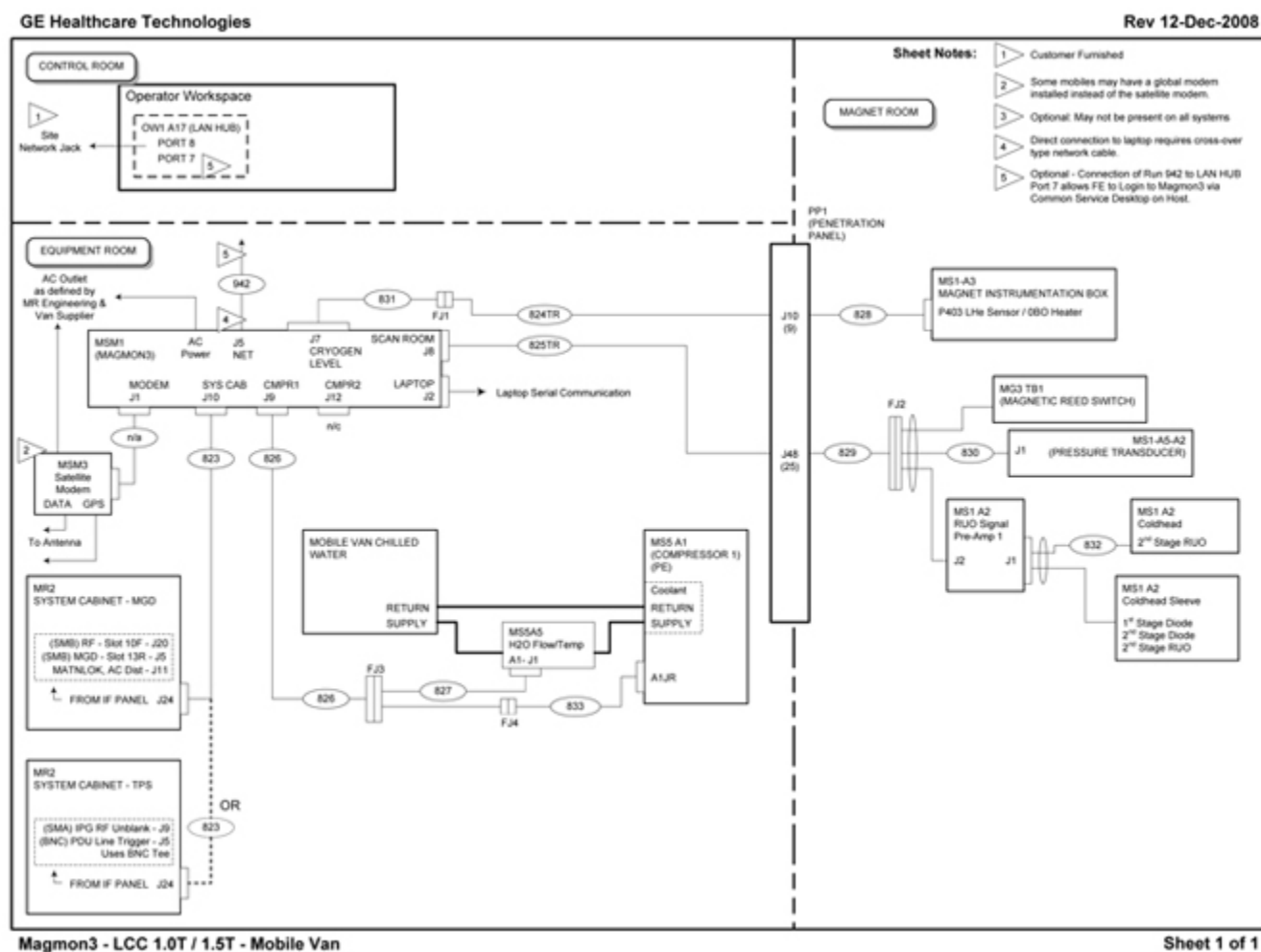
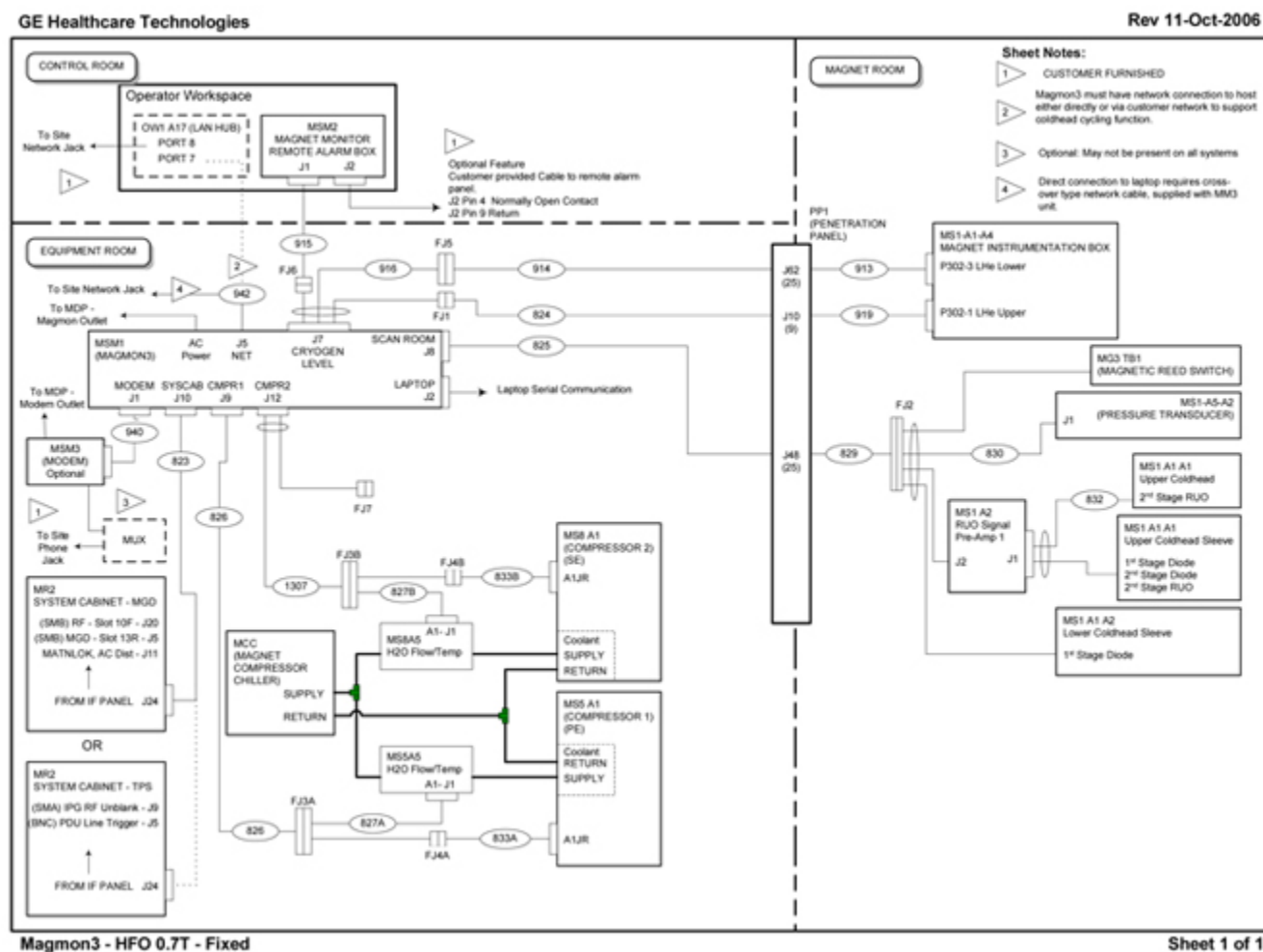


Illustration 4-2: LCC 1.0T/1.5T Mobile Van



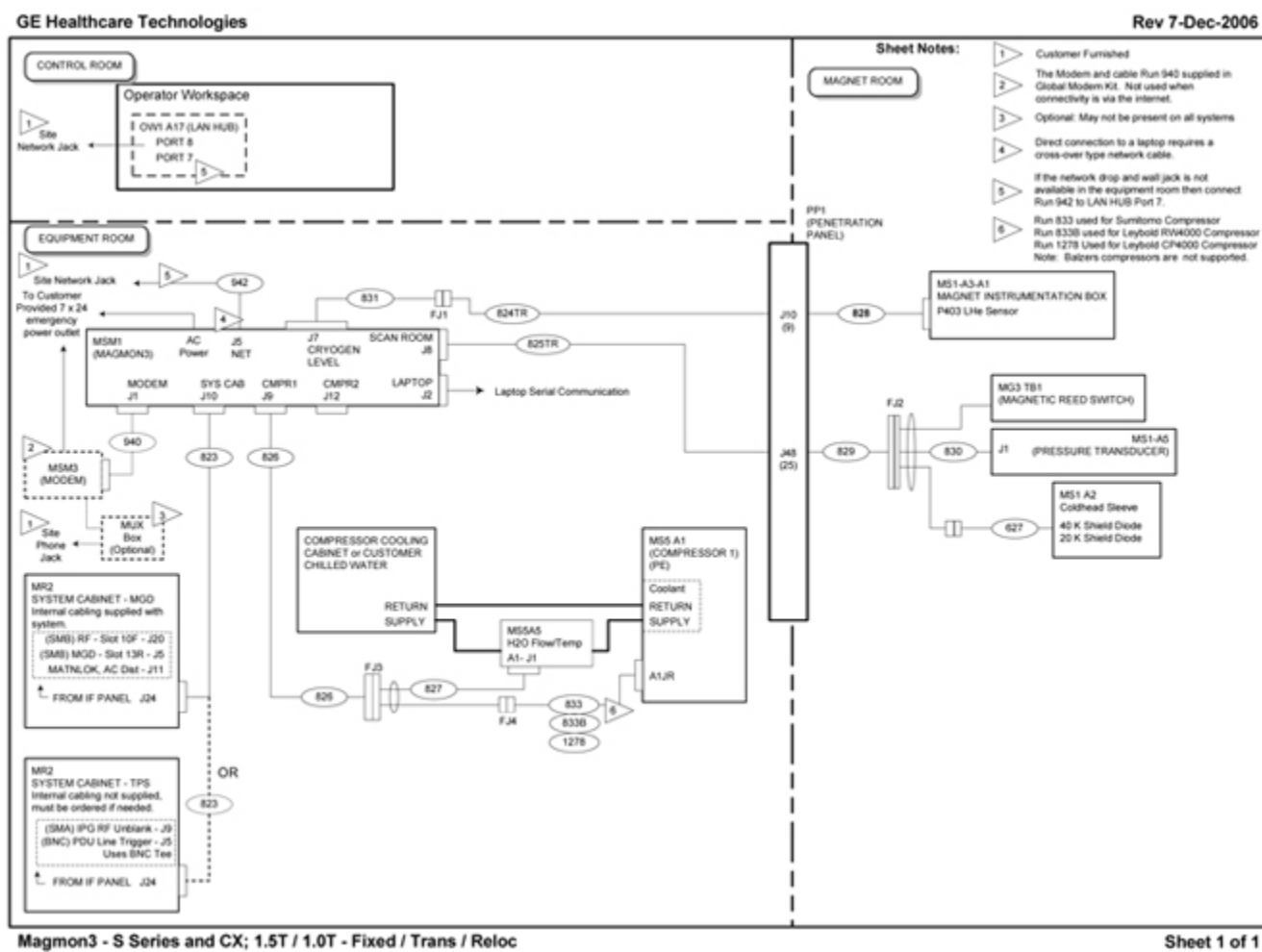
1.3 HFO 0.7T - Fixed

Illustration 4-3: HFO 0.7T - Fixed



1.4 S Series and CX - 1.5T/1.0T Fixed/Transportable/Relocatable

Illustration 4-4: S Series and CX - 1.5T/1.0T Fixed/Transportable/Relocatable



1.5 SX and CX - 1.0T/1.5T Mobile Van

Illustration 4-5: SX and CX - 1.0T/1.5T Mobile Van

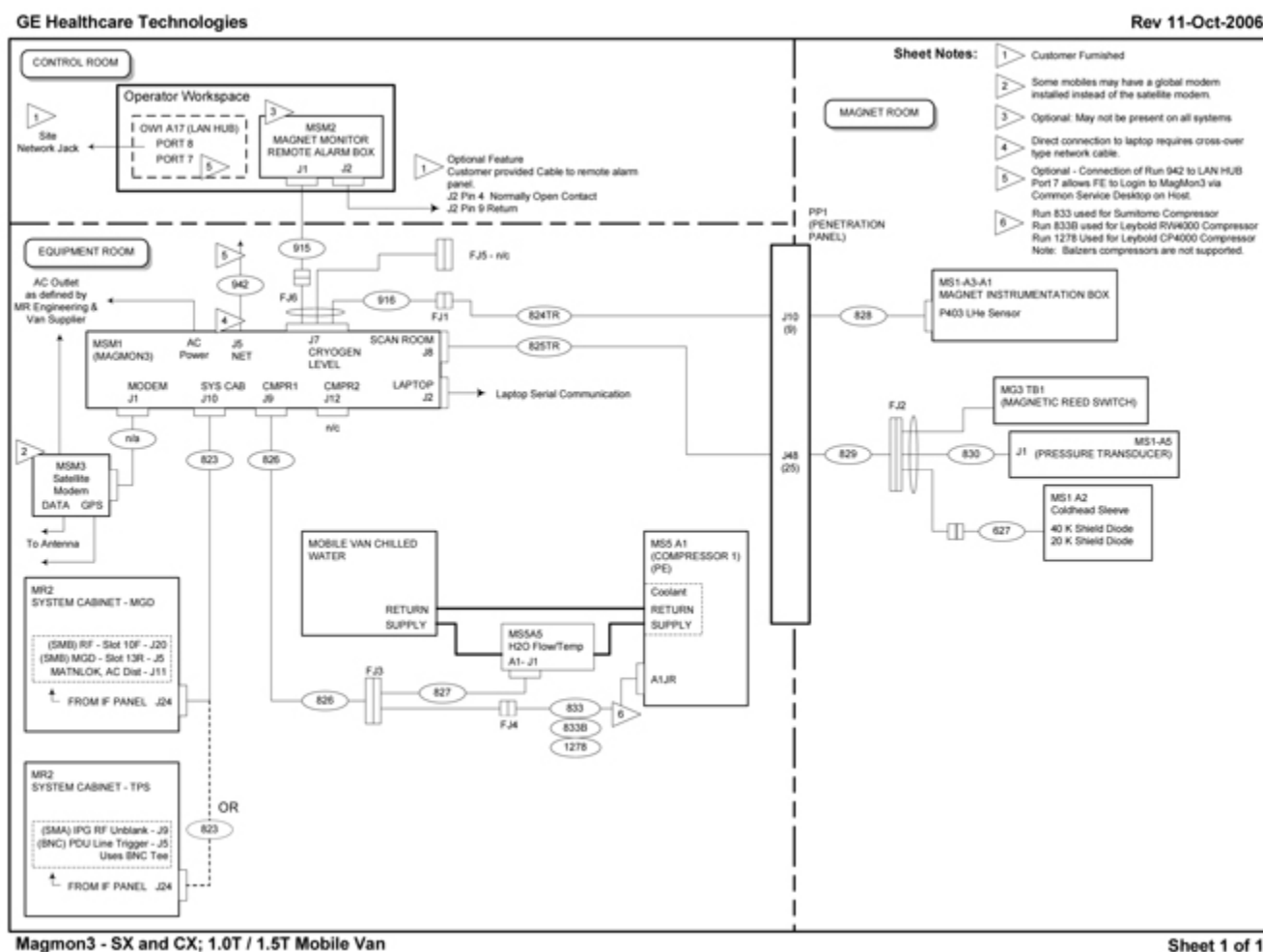
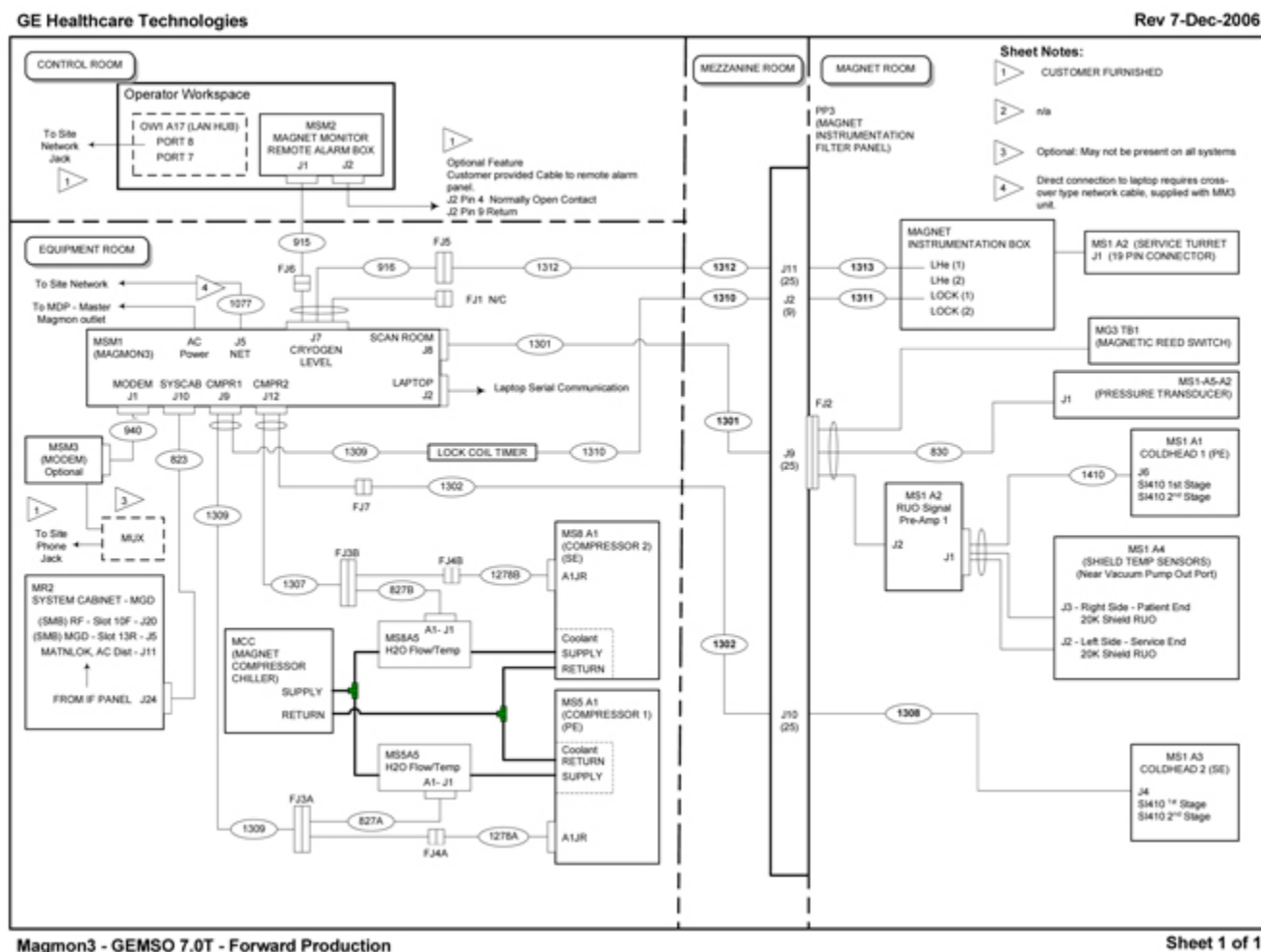
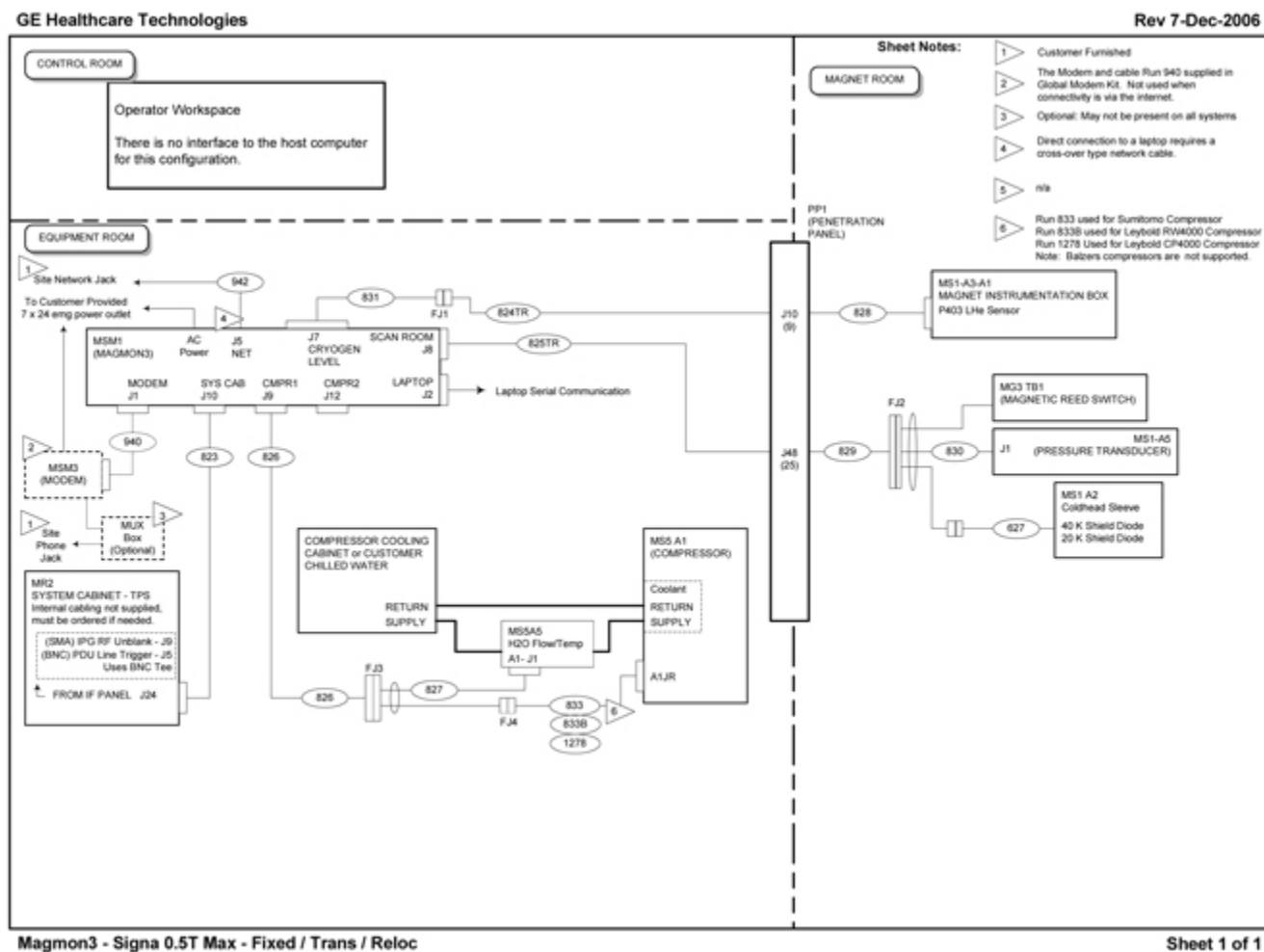


Illustration 4-6: GEMSO 7.0T - Forward Production



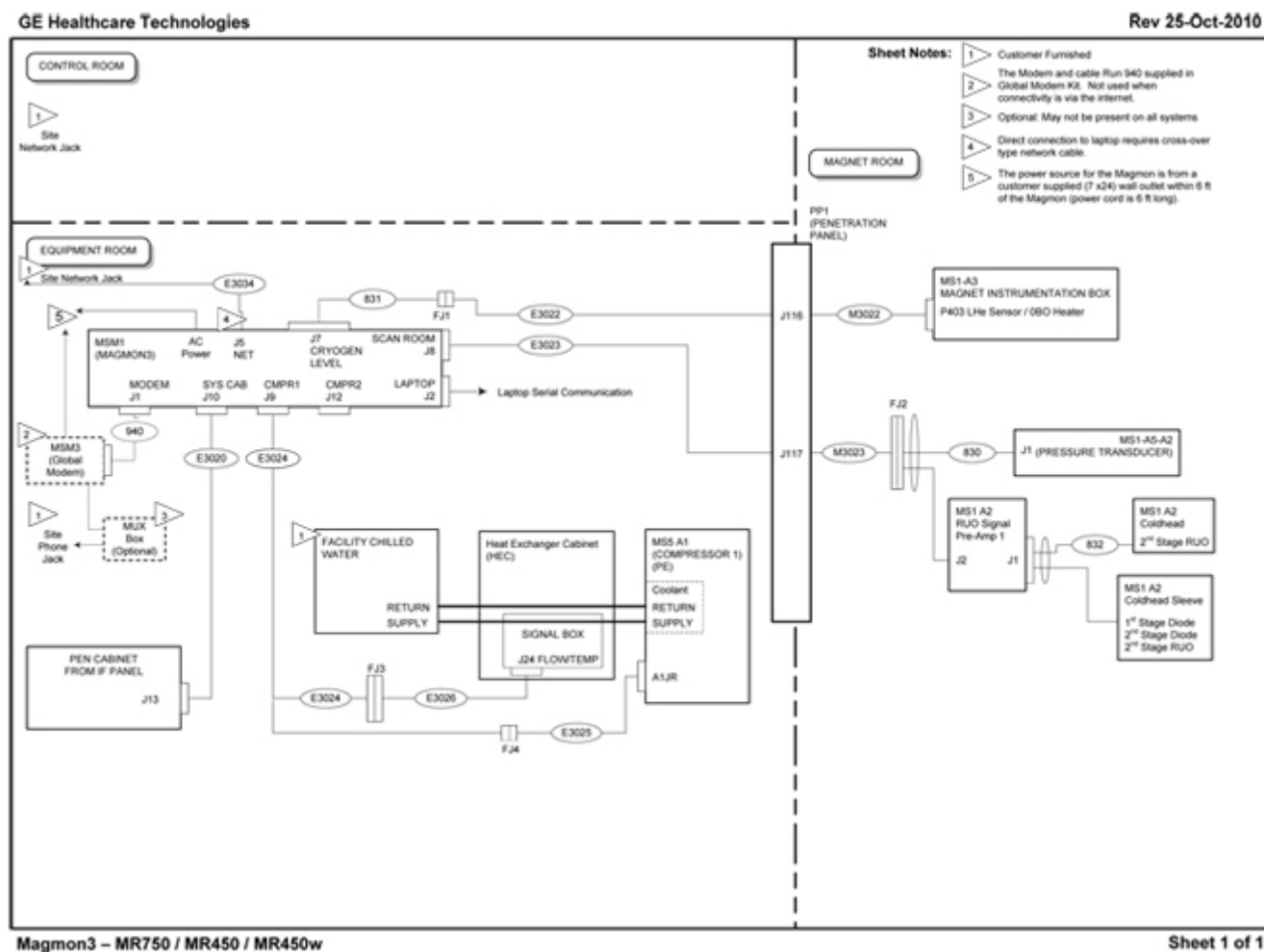
1.7 Signa 0.5T Max - Fixed/Transportable/Relocatable

Illustration 4-7: Signa 0.5T Max - Fixed/Transportable/Relocatable



1.8 Discovery MR750/MR450 and Optima MR450w

Illustration 4-8: Discovery MR750/MR450 and Optima MR450w



Chapter 5 Front Panel User Interface

1 Magmon3 Front Panel User Interface

1.1 Introduction

The Front Panel of the Magmon3 unit ([Illustration 5-1](#)) is made up of the following elements:

- LEDs showing AC Power, Heater activity, and Alarm activity.
- LCD Display for User Interaction.
- 16-button soft keypad.

This user interface provides easy access and viewing of the software revision, date/time, real-time sensor data and fault codes. Other features include the setting of fill modes and service modes through predefined access codes.

Illustration 5-1: Magmon3 Front Panel User Interface



1.2 Using the Magmon3 Front Panel

1.2.1 Softkey Buttons

1.2.1.1 Data

Pressing **Data** causes the Magmon3 unit to display a list of sensor parameters and real-time data. Press **Up** and **Down** to scroll through the list of items. A single item can be displayed indefinitely, and the Magmon3 will update that value (except for the He Level) once per minute.

NOTE: Leaving the **He Level** parameter on the screen will NOT force the Magmon3 to sample a new He Level value each minute. You must press **Fill Mode** to generate a new helium level sample each minute. See [Section 1.2.2.2](#).

NOTE: Cycling through the Data button entries will display the current IP address and InSite2 connection status.

1.2.1.2 Alarms

Pressing **Alarms** causes the Magmon3 to display a list of current fault codes. Press **Up** and **Down** to scroll through the list of items. You can only update items in this list by exiting from the Alarm menu and then re-entering the menu. The Alarm LED is lit if there are any fault codes to view.

NOTE: The Magmon3 tracks potential faults over a ten-minute period before the software activates the Alarm LED and updates the fault codes to minimize false reporting due to noisy transients.

1.2.1.3 Sample

Pressing **Sample** causes the Magmon3 to initiate a Helium Level sample and read the current helium level. It may take up to 90 seconds to update the display for the new level(s).

1.2.1.4 Home

Pressing **Home** puts the display into normal operating mode. In this mode, the display will cycle through the following information:

Table 5-1: Home Display

Screen	Display
1	Date Software Revision Time Number of Alarms
2	He Level <i>x.xx%</i>
3	He Pressure <i>x.xxx psi</i>

When you complete another operation, it is best to return to the Home display.

1.2.1.5 Call InSite

Pressing **Call InSite** causes the Magmon3 to immediately connect to InSite and upload any unsent data. Press **Yes** to confirm the action.

NOTE: This is only applicable to systems that are remotely monitored while on a GE service contract.

1.2.2 Fill Mode

Pressing **Fill Mode** allows the user to select one of two options: **Pre-fill Mode** or **Fill Mode**.

1.2.2.1 Pre-Fill Mode

The Pre-fill mode is used for LCC-type magnets prior to a helium fill. This option will change the pressure control settings that are used for heater cycling. The end result is the magnet's helium vessel pressure is reduced from 4 psi down to approximately 1.0 psi as long as the Cryogen Cooler Cabinet coldhead and recondensor are functioning normally.

NOTE: Pre-Fill Mode will run for up to 7 days.

If the user selects the Fill mode option before the time period expires, Pre-fill mode is automatically exited and the Fill mode functions take over.

If the time period expires or the user intentionally exits the Pre-fill mode via the soft keypad, the software will automatically put heater cycling into a post-fill operation. During post-fill the operating parameters for pressure control are changed and the heater will be activated for an extended period of time, allowing the magnet's helium vessel pressure to return to its normal operating state. When in post-fill, heater cycling will not activate any heater alarms for 48 hours.

NOTE: Pre-fill mode should only be used for LCC, LCC-RB, DVw, and LCC300 magnet types.

1.2.2.2 Fill Mode

The Fill mode performs the following functions:

1. Exits Pre-fill mode (if necessary)
2. Disables the pressure control function for heater cycling (for LCC and HFO magnets).
3. Initiates a helium level sample once every minute and refreshes the LCD display information.

The user must select and initiate Fill mode before inserting the fill line stinger into the magnet. When the filling operation is complete, the user will exit the Fill mode option and the software will automatically go into a post-fill operation. During post-fill, the operating parameters for pressure control are changed and the heater will be activated for an extended period of time, allowing the magnet's helium vessel pressure to return to its normal operating state. When in post-fill, heater cycling will not activate any heater alarms for 48 hours.

1.2.3 Service Mode

Pressing **Service Mode** allows the user to select one of these options:

- Web Mode for configuration setup of the unit, using either a DHCP address (**Same IP**) or a **Fixed IP** address.
- Service Mode (for GE Service only).

1.2.3.1 Web Mode

Users must put the Magmon3 into Web mode to use the configuration tool's web interface. Any user, whether a GE employee or not, can connect and log in to the interface with a laptop or the system host computer. Once the user is logged in, the configuration settings of the Magmon3 can be changed and saved.

When you press **Service Mode**, you must know what form of Web Mode to enter. Pressing **Service Mode** repeatedly will cycle the display through the options:

- **Web Mode - Same IP:** Magmon3 connected to a network and configured as DHCP or a pre-configured static IP address.
- **Web Mode - Fixed IP:** Enables the web interface and forces the Magmon3 IP address to 192.168.0.1.

Press **Yes** when the correct option is displayed to place the Magmon3 in Web Mode. No password entry is required.

NOTE: The unit will remain in Web Mode for 45 minutes, after which time it will terminate the connection. Web Mode will also be terminated if the AC power to the unit is cycled off, then on.

Once you are in Web Mode, press **Data** repeatedly until the IP address is displayed. Write the address down, as you will need it when setting up your laptop or host computer to connect to the unit.

Once you are in Web Mode, you can now attempt to connect to the unit. Refer to [Chapter 6, Connecting a Laptop to the Magmon3](#).

Chapter 6 Laptop Connection

1 Connecting to Magmon3 with a Laptop

1.1 Connecting Via The Building Network

This procedure is used to log into Magmon from the Service Browser on the customer's scanner, or from a laptop attached to the customer's network.

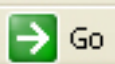
1.1.1 Prerequisites

- Magmon3 was previously configured for network operation using a direct connection between this device and a laptop.
- Magmon3 has been successfully communicating over the building network.

1.1.2 Precondition

Magmon3 is connected to the building network via Magmon3 – J5 port.

1.1.3 Procedure

1. Find the IP address assigned to the Magmon3 by pressing **Data**, then **Up** repeatedly until the IP Address displays.
2. Write down the IP Address for use later.
3. Press **Service Mode** repeatedly until the display shows **Start Web Mode? (Yes/No) Same IP**. Press **Yes** to activate Web Mode.
4. Connect the laptop to the building network using a CAT5-type network cable.
5. Open a web browser session on the laptop.
6. Type the Magmon3's IP address (recorded earlier) into the laptop browser's address bar, and click  (Go). The browser should display the Magmon3 *Service Login* screen.
7. Go to [Chapter 7, Login Modes](#) for further instructions.

1.2 Direct Network Cable Connection to Laptop and Magmon3

Use this procedure for these conditions:

- First-time setup of the Magmon3.

1.2.1 Precondition

Laptop connected directly to Magmon3 – J5 port using a crossover CAT5-type network cable.

1.2.2 Procedure

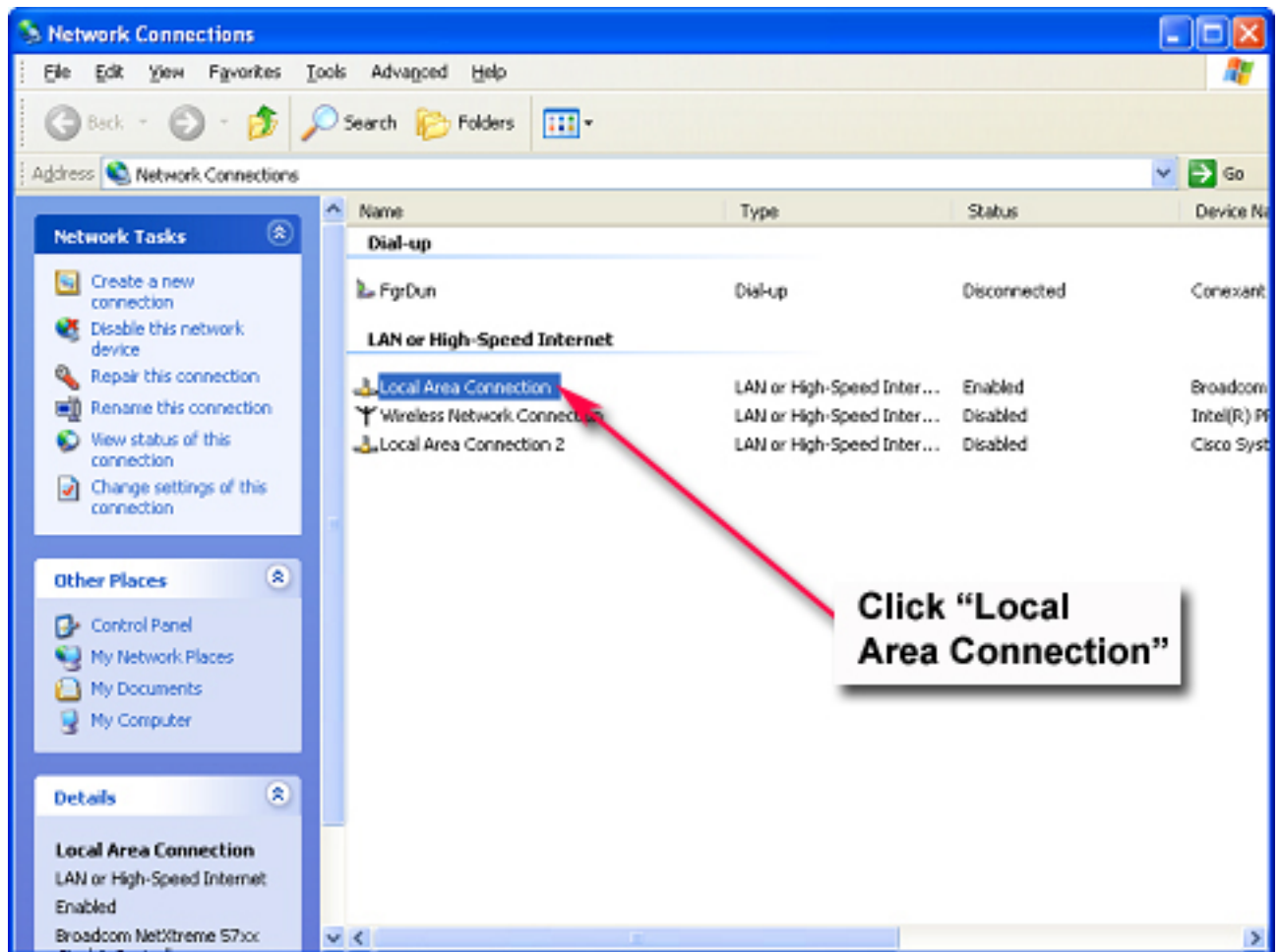
1.



Right-click on (My Network Places) on the laptop's desktop.

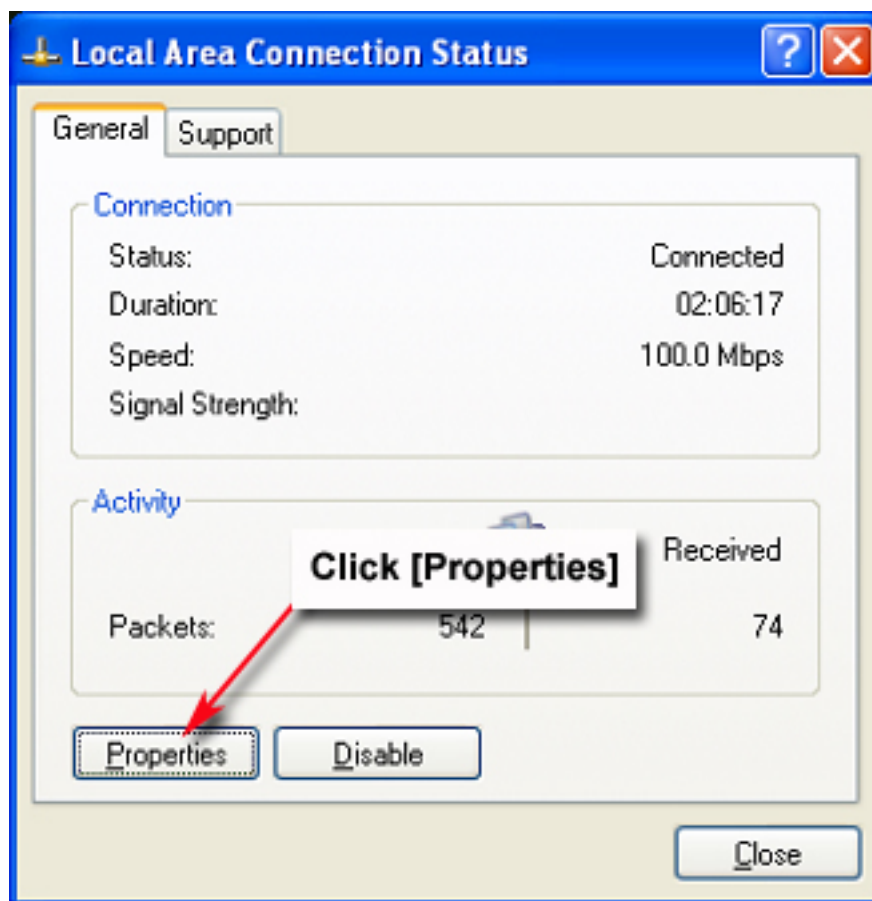
2. Select **Properties** from the menu. The *Network Connections* window (Illustration 6-1) will open.

Illustration 6-1: Network Connections Window



3. Disable the wireless network connection (if enabled). Right-click on **Wireless Network Connection**, and select [Disable].
4. Double-click **Local Area Connection**. The *Local Area Connection Status* window (Illustration 6-2) will open.

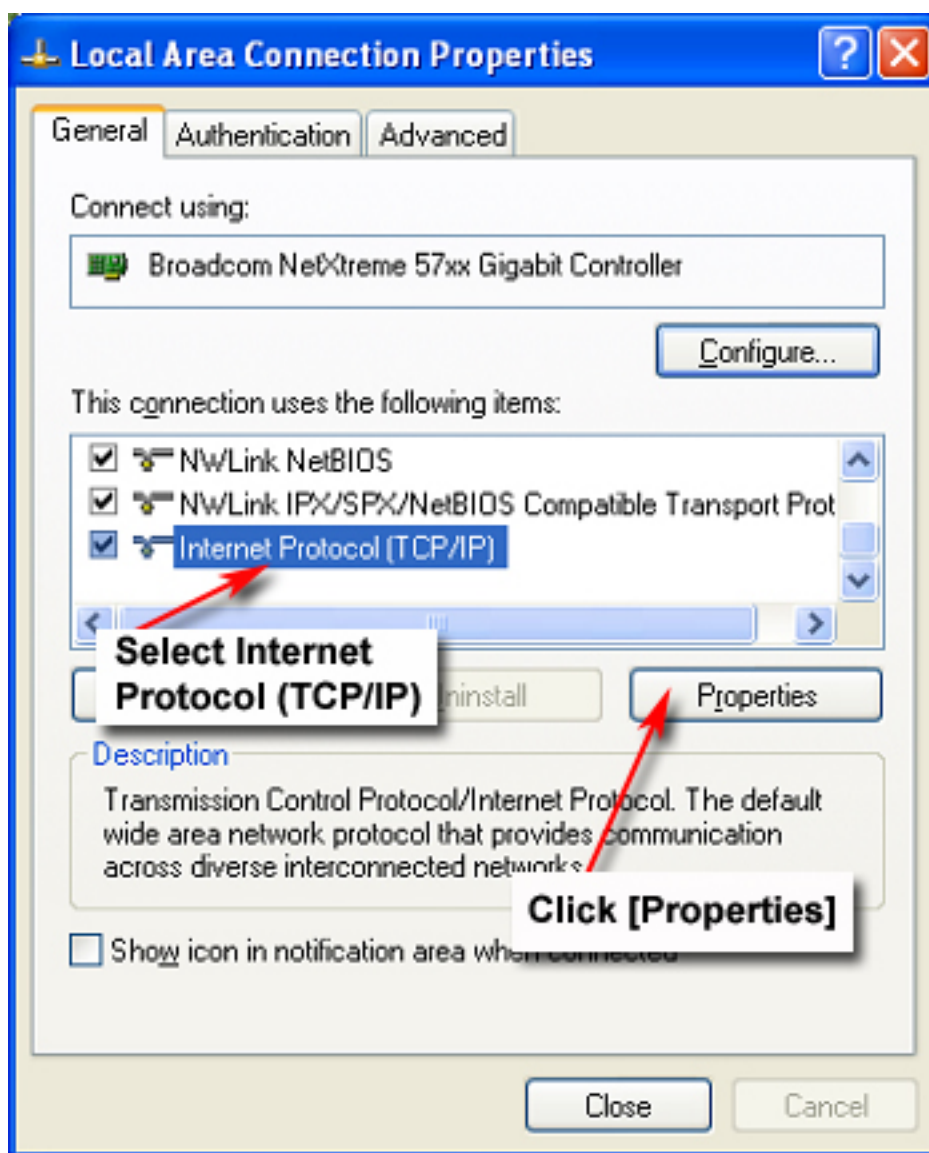
Illustration 6-2: Local Area Connection Status Window



NOTE: Depending on the laptop's configuration, the *Local Area Connection Properties* window (Illustration 6-3) may be display. In this case, skip to Step 6.

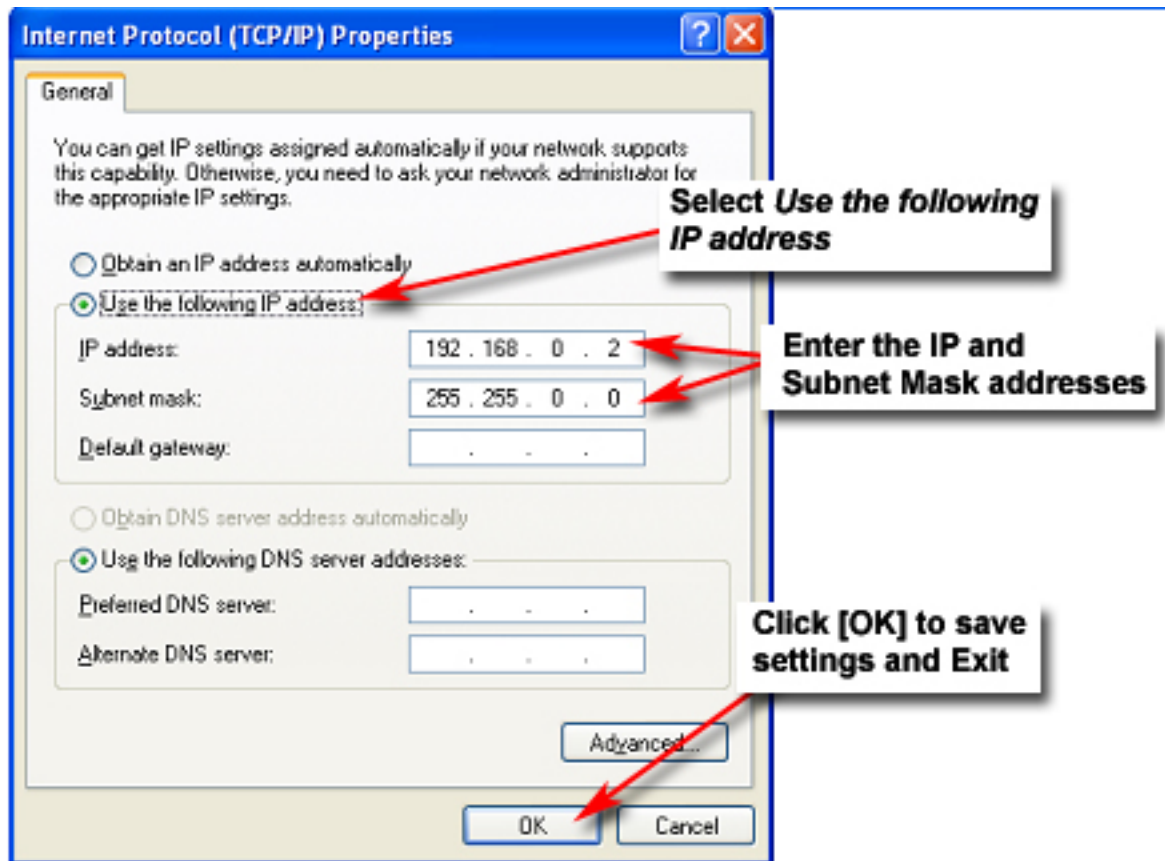
5. Click [Properties] to open the *Local Area Connection Properties* window shown below.
6. Select **Internet Protocol (TCP/IP)**, and click [Properties]. The *Internet Protocol Properties* window (Illustration 6-4) will open.

Illustration 6-3: Local Area Connection Properties Window



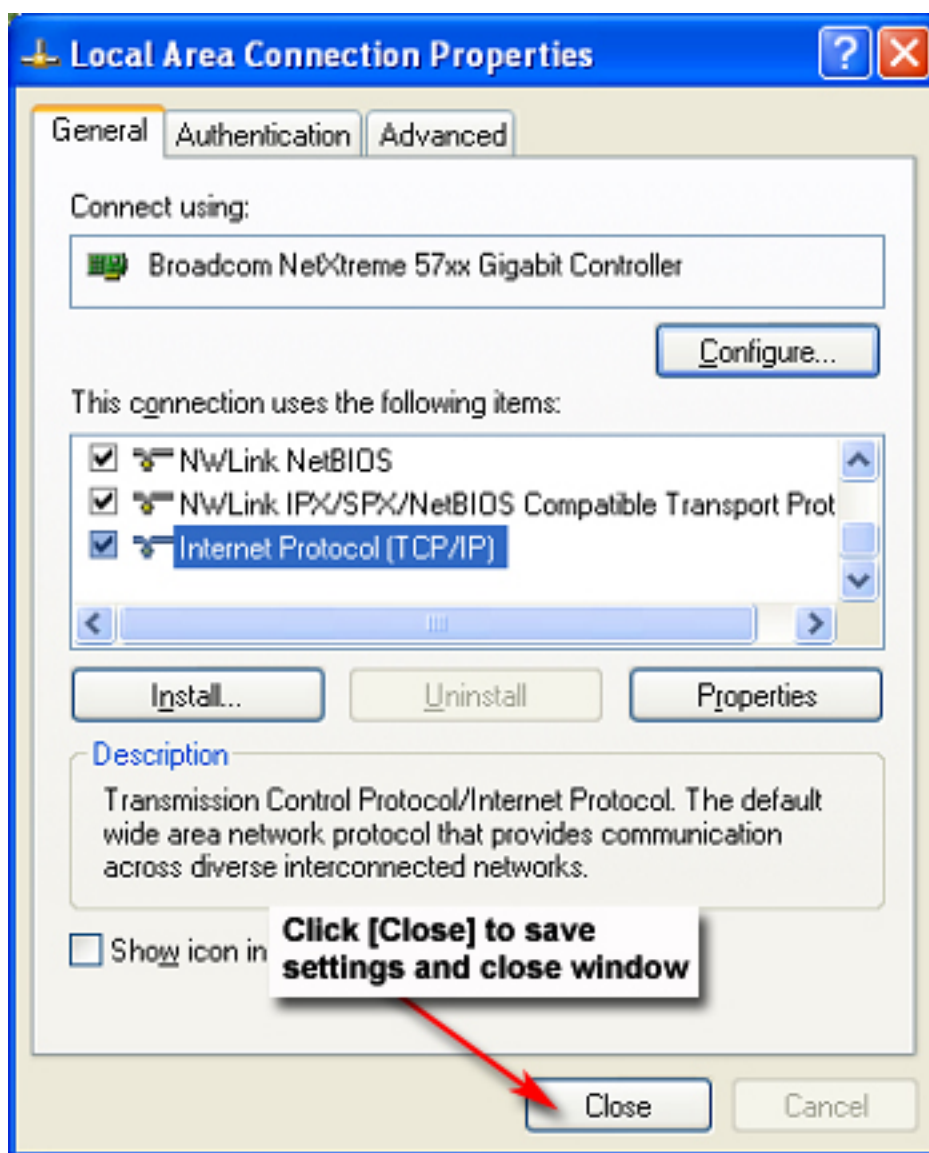
7. Click **Use the following IP address** on the *Internet Protocol (TCP/IP) Properties* window.

Illustration 6-4: Internet Protocol (TCP/IP) Properties Window



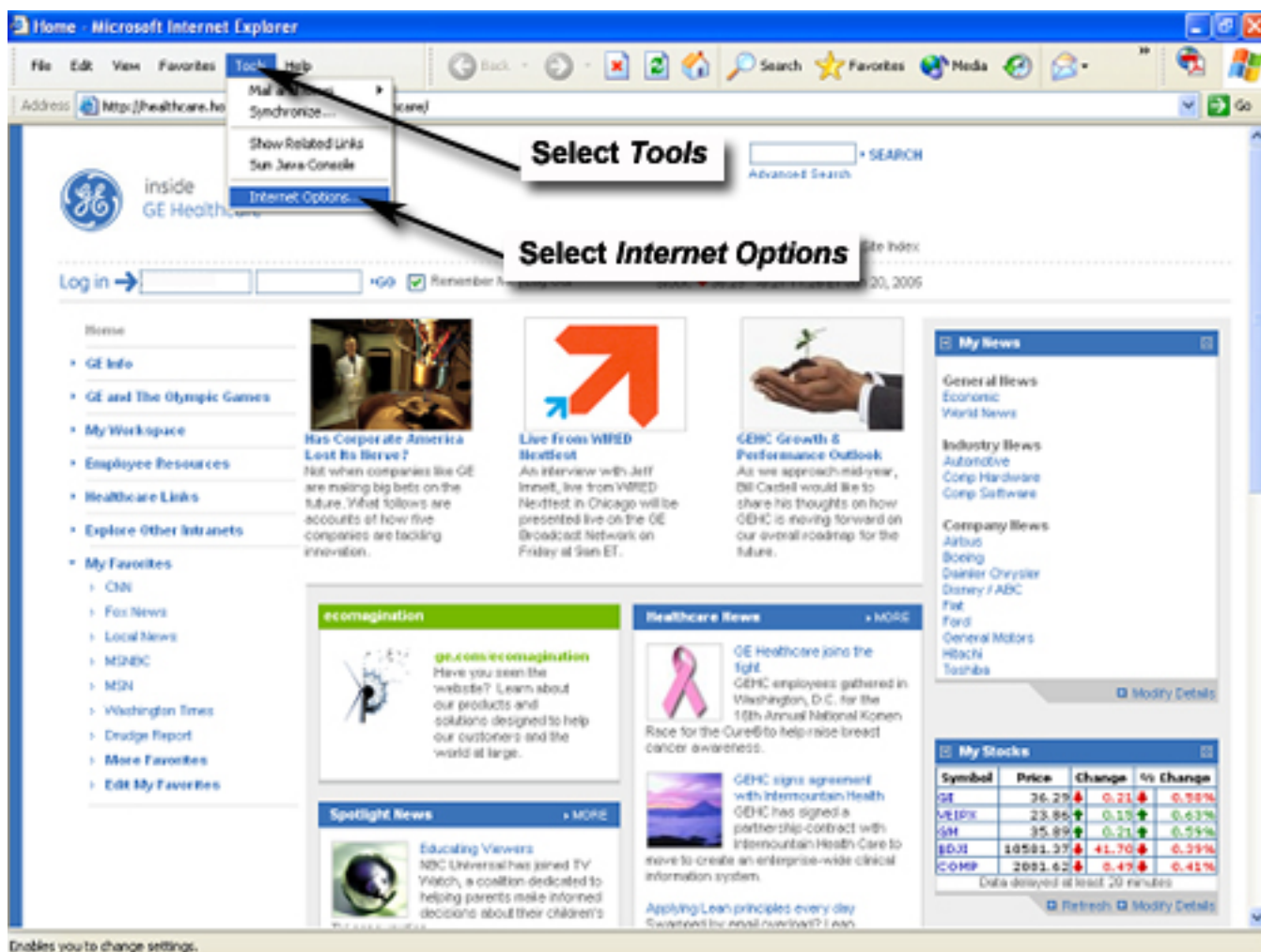
8. Enter the following addresses as shown in [Illustration 6-4](#):
 - Type **192 . 168 . 0 . 2** in the *IP address* box.
 - Type **255 . 255 . 0 . 0** in the *Subnet mask* box.
9. Click [OK] to save settings and exit.
10. Click [OK] or [Close] to save the new Local Area Connection Properties ([Illustration 6-5](#)). This may take a moment. Make sure to close the *Local Area Connection Status* window.

Illustration 6-5: Local Area Connection Properties Window



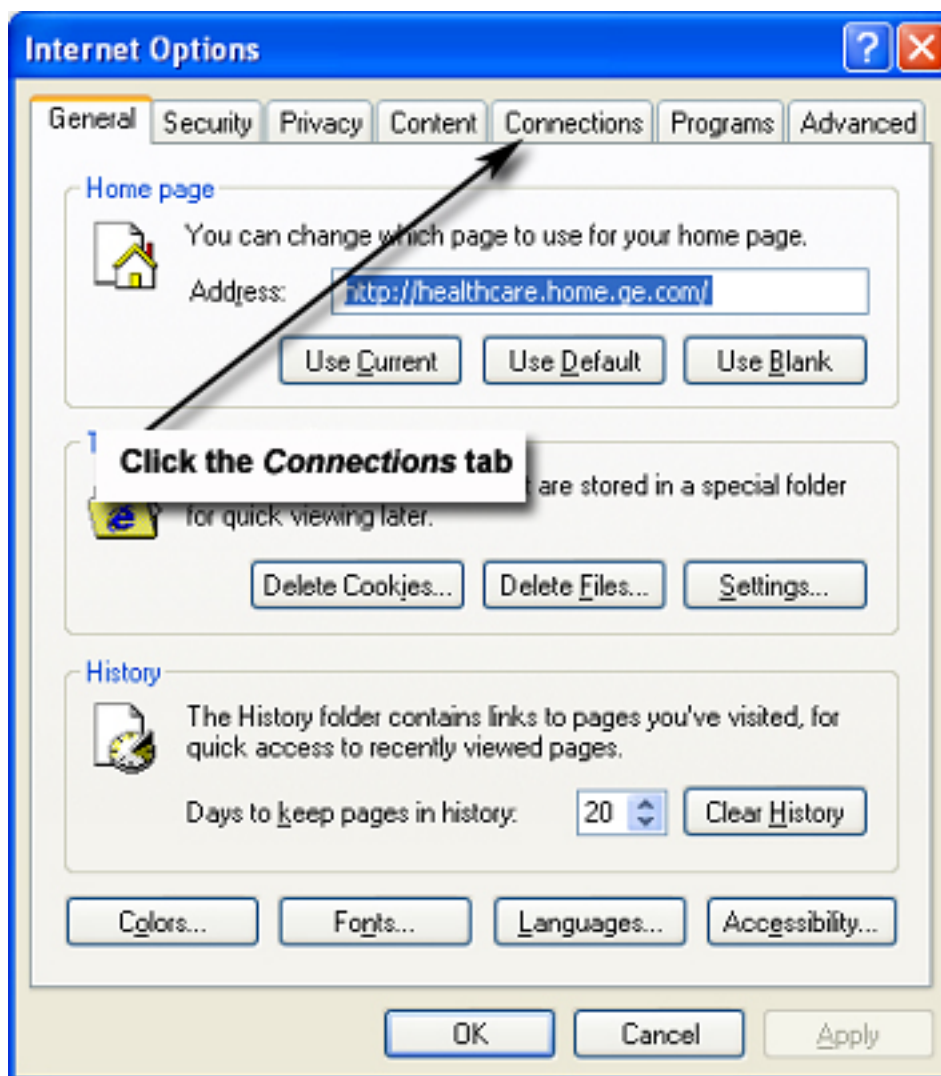
11. Launch Internet Explorer from the laptop's desktop.
12. In Internet Explorer, select Tools→Internet Options as shown below.

Illustration 6-6: Internet Explorer Menu Selections



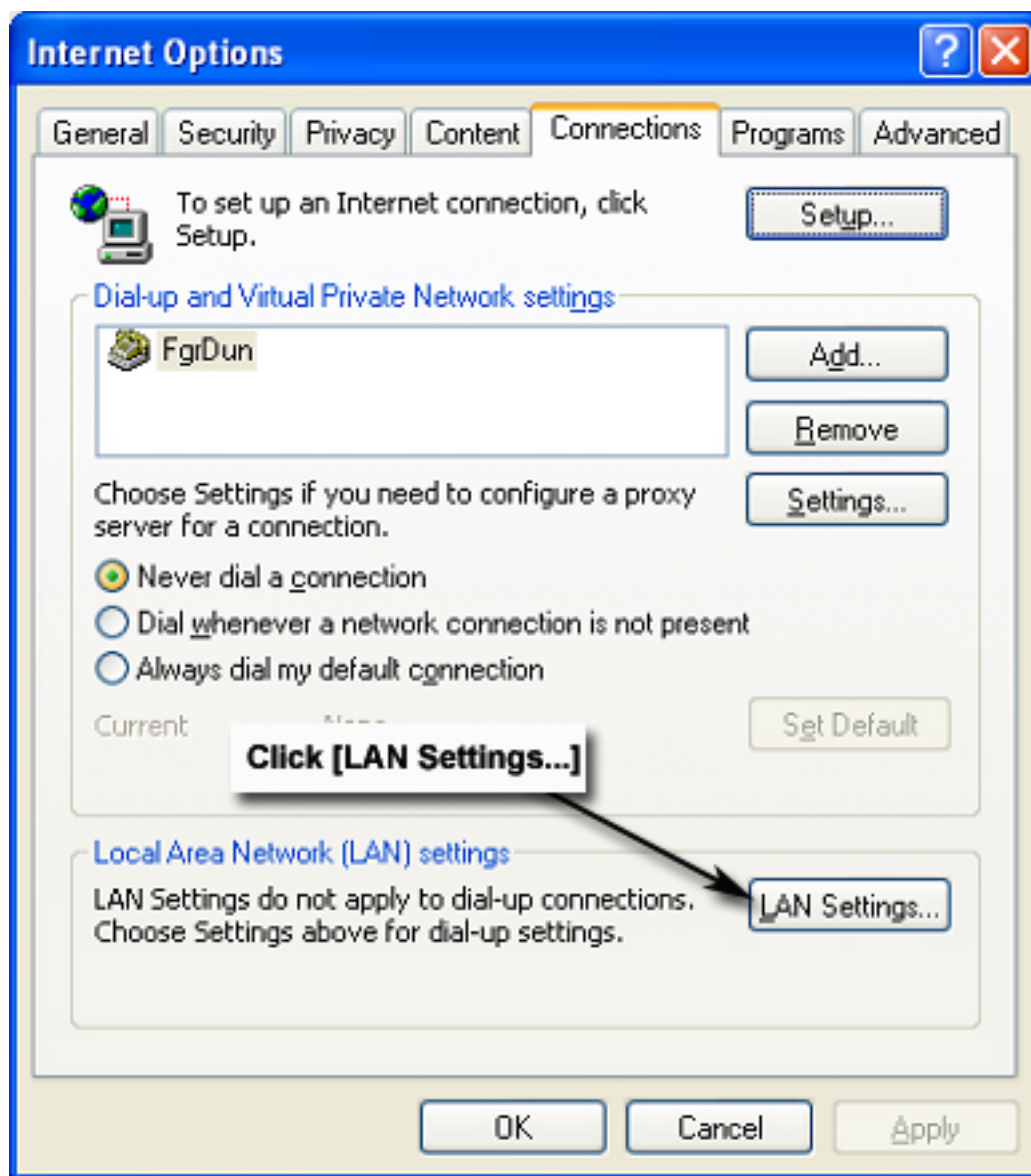
13. In the *Internet Options* window, click the **Connections** tab.

Illustration 6-7: Internet Options Window



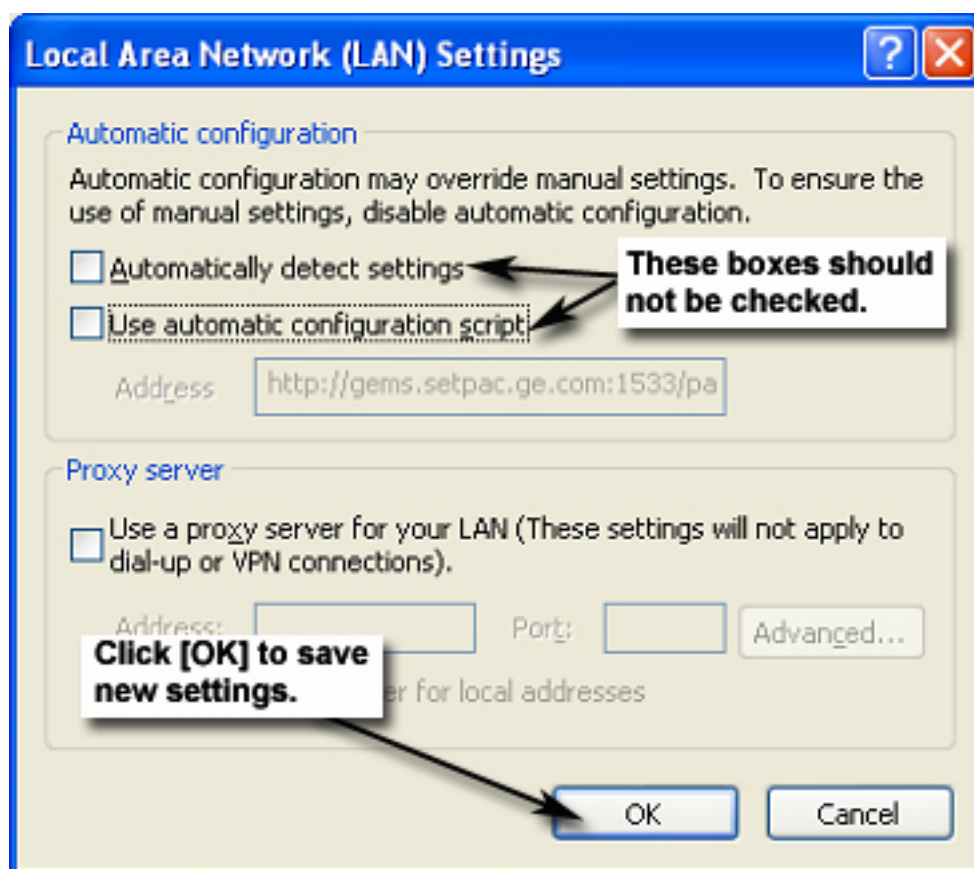
14. Click [LAN Settings] on the *Internet Options* window.

Illustration 6-8: Internet Options Connection Tab



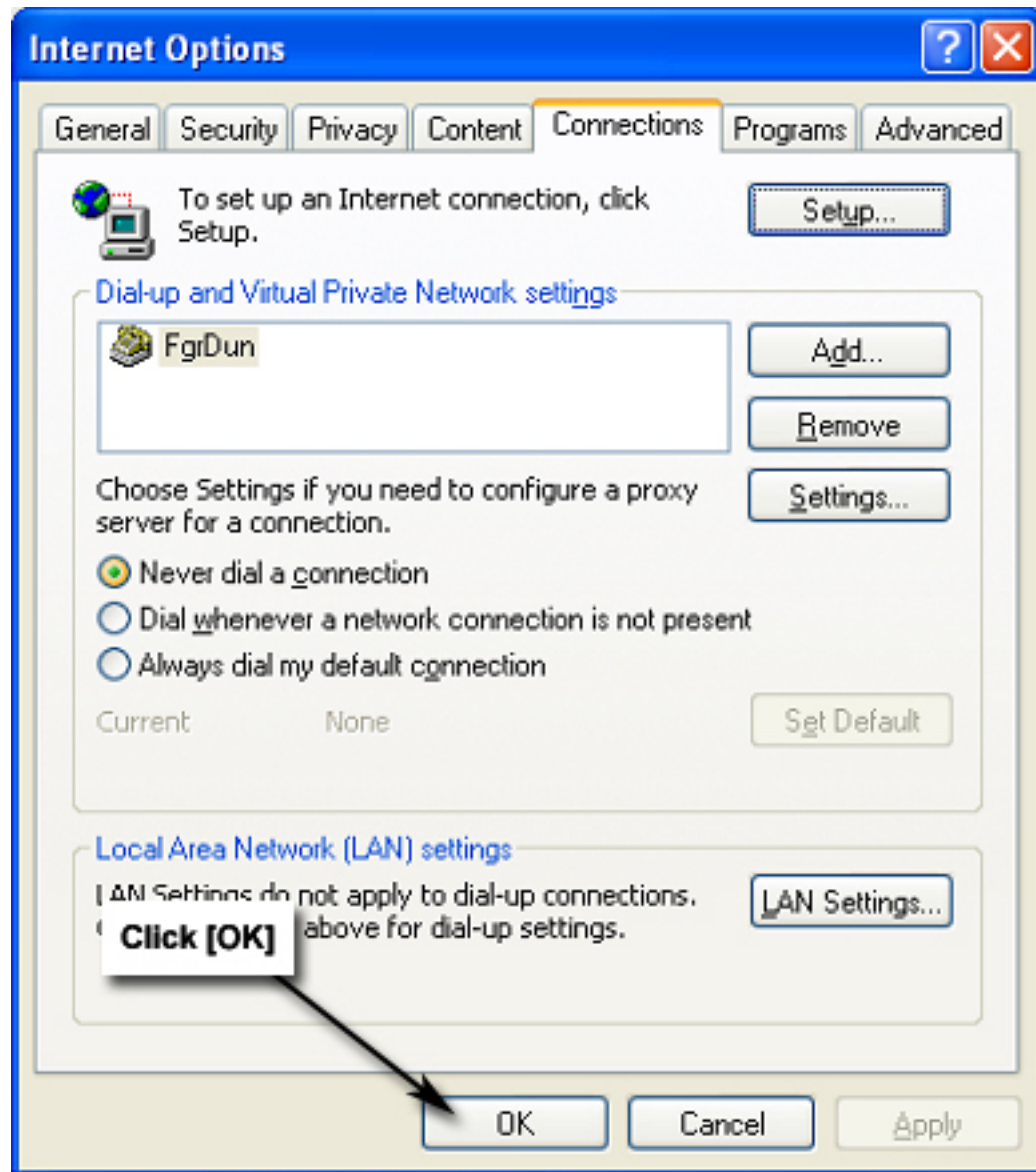
15. When the *Local Area Network (LAN) Settings* window opens, make sure the *Automatically detect settings* and *Use automatic configuration script* boxes are **not** selected.

Illustration 6-9: Local Area Network (LAN) Settings Window



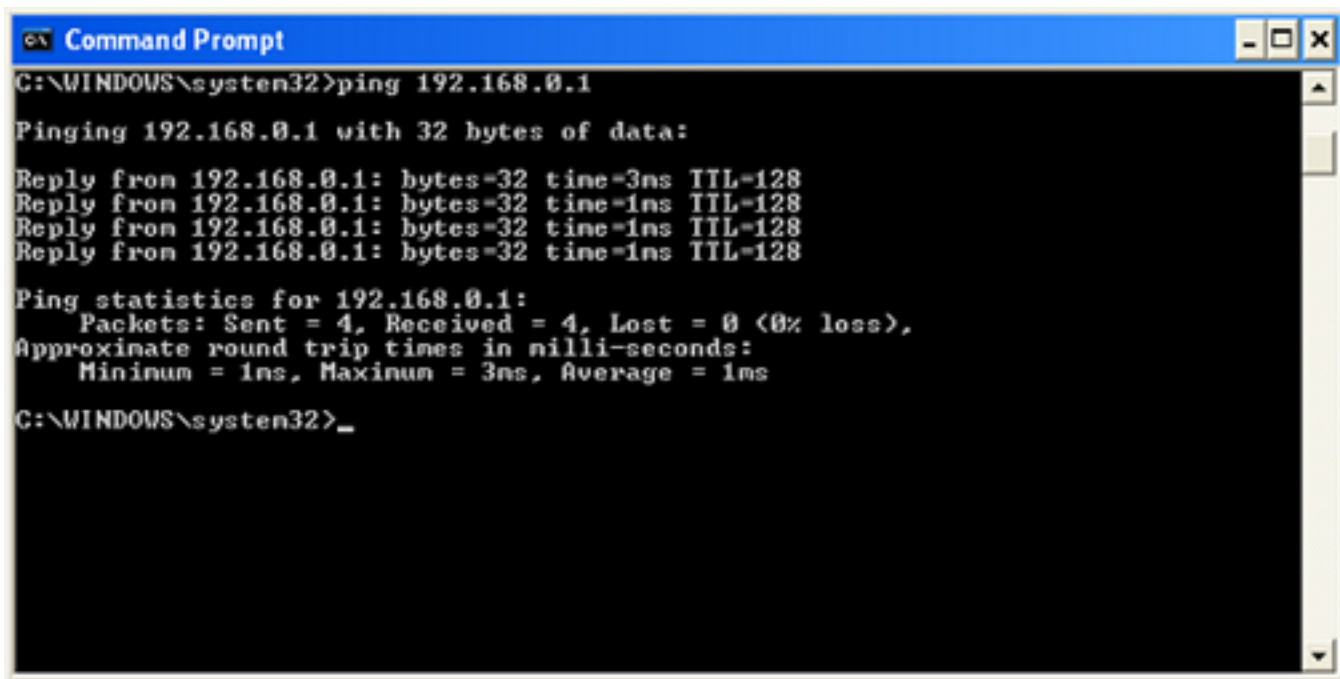
16. Click [OK] to save the LAN Settings. Click [OK] to save all settings and close the *Internet Options* window.

Illustration 6-10: Internet Options Window



17. On the laptop, open a *Command Prompt* window. In Windows XP, click the  (c:\) icon in the taskbar. (Another option is to go to Start→Programs→Accessories→Command Prompt.)
18. Type **ping 192.168.0.1**<Enter>. If the Magmon3 unit responds ([Illustration 6-11](#)), close the *Command Prompt* window, and proceed to the next step. If not, recheck your settings.

Illustration 6-11: Command Prompt Window

A screenshot of a Windows Command Prompt window. The title bar is blue and says "Command Prompt". The window has a black background with white text. The text shows a command to ping 192.168.0.1, followed by four successful replies with 32 bytes of data, 1ms time, and TTL=128. Ping statistics show 4 packets sent, 4 received, 0 lost, and an average round trip time of 1ms.

```
C:\WINDOWS\system32>ping 192.168.0.1

Pinging 192.168.0.1 with 32 bytes of data:

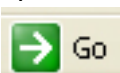
Reply from 192.168.0.1: bytes=32 time=3ms TTL=128
Reply from 192.168.0.1: bytes=32 time=1ms TTL=128
Reply from 192.168.0.1: bytes=32 time=1ms TTL=128
Reply from 192.168.0.1: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 3ms, Average = 1ms

C:\WINDOWS\system32>_
```

19. Return to Internet Explorer and enter the Magmon3's IP address (**192.168.0.1**) in the

address field. Click



(Go). The browser should display the *Magmon3 Service Login* screen.

Illustration 6-12: Magmon3 Service Login Screen

The screenshot shows a web browser window titled "Magmon3 Proprietary Web Interface - Microsoft Internet Explorer". The address bar displays "192.168.0.1". The page has a light yellow background. At the top left is the "GE Healthcare" logo, and at the top right is the word "Login". Below the logo is a horizontal line. The main heading is "Magmon3 Service Login", followed by the instruction "Enter a Username and Password to proceed." There are two input fields: "Username:" and "Password:". Below these fields is a "Submit" button. The browser's status bar at the bottom shows "Done" and "Internet".

Magmon3 Proprietary Web Interface - Microsoft Internet Explorer

File Edit View Favorites Tools Help

Address 192.168.0.1

GE Healthcare Login

Magmon3 Service Login

Enter a Username and Password to proceed.

Username:

Password:

Done Internet

20. Go to [Chapter 7, Login Modes](#) for further instructions.

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Chapter 7 Login Information

1 Login Modes

1.1 Customer Access – Logging Into Magmon3

All Magmon3 units are pre-configured by the manufacturer in non-Proprietary mode before the units ship to GE. Customers not on a GE Service contract will need access to the unit to be able to perform some configuration tasks in the event of a hardware failure where the Magmon3 would need replacement.

1.1.1 *Magmon3 Non-Proprietary Mode*

The unit can perform these basic functions in non-Proprietary mode:

- Display last sample of **He Level (%)**.
- Allow the user to initiate **He Level Sample** to update the displayed information.
- Display **He Vessel Pressure (psi)**.
- Perform pressure control for magnets with Recondenser Refrigeration systems.
- Display alarms when a monitored parameter is out of range.
- For Signa OpenSpeed products, the Magmon3 will communicate with the MRI System Host PC via a network connection, and provide parameter data needed to support coldhead cycling.
- Allows the user to access **Data Mode** for viewing current sensor values.
- Allows the user to access **Fill Mode** to increase He Level sampling during the fill operation.

1.1.2 *Preconditions*

1. Laptop directly connected to the Magmon3 using either a network crossover cable or a serial cable.
2. Laptop communication configured to support either network or serial connection.
3. Laptop communicating with the Magmon3 unit and the login screen displayed on the laptop's web browser.

1.1.3 *Logging In*

At the Login screen, enter this information, as in [Illustration 7-1](#) (input is case sensitive):

Username: **MMConfig**

Password: **ConfigMonitor**

Illustration 7-1: Magmon3 Service Login Screen

GE Healthcare

Logi

Magmon3 Service Login

Enter a Username and Password to proceed.

Username: MMConfig

Password: ■ ■ ■ ■ ■

Submit

Done

NOTE: The Magmon3 will disconnect users after 45 minutes.

Magnet Monitor type must be selected. Refer to [Chapter 9, Configuration Setup](#) for more information.

Chapter 8 Overview of Web Page

1 Magmon3 Web Page Screen Overview

NOTE: The web screens and detailed information is provided in the order that it appears on the Navigation menu on the left side of the screen.

1.1 Basic Web Page Layout

This web page shows the basic layout of the Magmon3 web interface. The left column is a navigation menu where you can select one of the available web pages. Selecting one of the items in the navigation menu changes the page shown in the main section. Upon login, the main section will display the *Version Info* screen.

Illustration 8-1: Basic Web Page Layout

The screenshot displays the Magmon3 Web Interface. On the left is a navigation menu with the following items: 'Log Out', 'Configuration', 'Set Time/Date', 'Magnet Type', 'Network', and a 'goahead WEB SERVER' logo. The main content area on the right is titled 'GE Healthcare' and 'Version Info'. It contains a table with two columns: 'Component' and 'Version'.

Component	Version
System Version:	1.51
FPGA Revision:	01h
Main Program Version:	1.51
Web Server Version:	1.15
InSite2 Version:	
Magmon HW Serial Number:	ACN6290391

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Chapter 9 Configuration

1 Configuration Setup

1.1 Set Time and Date

The *Set Time/Date* page allows the setting of the internal time and date for the Magmon. This process must be done via a laptop connection, and cannot be done remotely. The internal time and date are used to timestamp the Minute Data and the Event Log items. The time zone setting is required to synchronize the MagMon3's time to the back office time.

NOTE: The Magmon3 does not automatically change settings when daylight savings time goes into effect.

See [Illustration 9-1](#) for details on the following steps.

1. Set the Date, Time, Month, Year, Hour, and Minutes by clicking on each drop-down menu and making a selection.
2. Set the Time Zone by selecting the time zone for your area from the drop-down menu. For example, if in Germany, select: GMT +1.

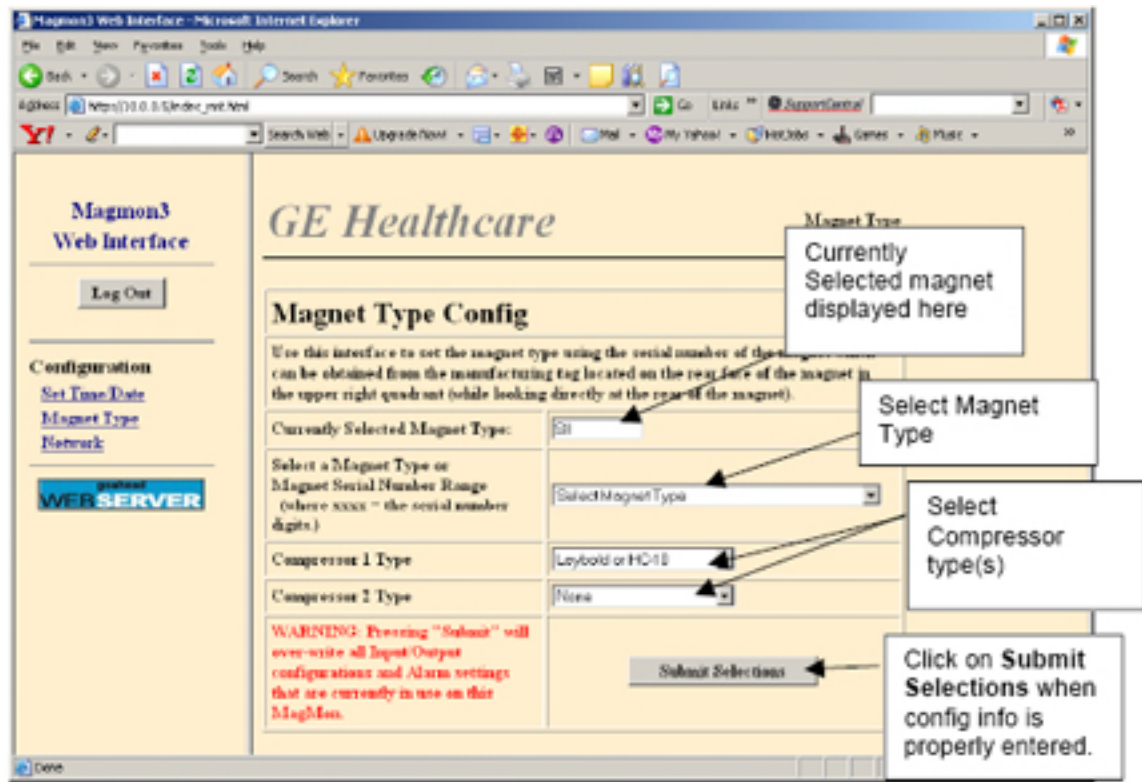
Illustration 9-1: Set Time and Date

1.2 Magnet Type

The *Magnet Type* page configures the Magmon3 to one of the predefined magnet types. The number of sensors, type of compressors, and alarm parameters will vary between magnet model types, so the magnet type selection is a key element in the operation of the magnet and magnet monitor.

1. Select the magnet type from the drop-down menu that most closely matches the system type that the Magmon3 will be monitoring. If not sure about what magnet type to select, contact the Magnet Support Team and your regional On Line Center.

Illustration 9-2: Magnet Type



2. Select the proper compressor type from the menu shown above. Compressor 2 type should only be used when two compressors are connected to the same magnet.
3. Click [Submit System Type] after identifying the magnet type.

1.3 Network IP Configuration

A network connection must be made between the Magmon3 and the High-Field Open (HFO) Host Computer's LAN Port 7. Go to [Chapter 12, Appendix, Section 1, Prerequisite Information for Non-Proprietary Operation](#) for network information on HFO Systems.

Refer to [Illustration 9-3](#). The following is the communication path for supporting the Coldhead Switching feature on the HFO system:

1. Click [Network].
2. Enter the information provided by the customer's network administrator for IP Configuration and DNS Configuration.
3. Click [Submit Changes].



NOTICE

On the Configure Network IP page, make sure the *No Network connection when not in Web Mode* box is NOT selected.

Illustration 9-3: IP Configuration Network Settings

1.4 Clear Data

NOTE: Using [Reset All Config Files] is useful when moving a Magmon3 from one system to another.

Illustration 9-4: Clear Data

GE Healthcare		Clear Data
Erase All Saved Data		
Pressing this button will erase all minute data and the event logs. This command is useful for removing bad data that does not need to be sent to the back office. For example, if the magmon was running for a long time without being connected to the magnet sensors, the magmon's internal 31-day data storage will be filled with useless data and error events. This command can be used to erase that data before the magmon is reconnected to the system and that data is uploaded to Insite. This command will not change any settings or configurations.		
WARNING: Pressing "Erase Data" will erase previously stored minute data and event logs.	Erase Data	
Return All Settings to Factory Defaults		
Use this interface to erase all configuration files and all saved data on this Magnet Monitor. Pressing this button will attempt to return the magnet Monitor to the condition it was in when it left the factory.		
WARNING: Pressing "Reset All Config Files" will erase all configuration settings and all saved data on this Magnet Monitor.	Reset All Config Files	

Chapter 10 Maintenance and Replacement

1 Maintenance and Replacement

The only maintenance action for Magmon3 is to check the time of day occasionally.

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Chapter 11 Renewal Parts

1 Renewal Parts

Table 11-1: Cables

Item	Part Number	Run #	Description	FRU
1	46-318042P1	627	FJ2 (Run 830) to MS1-A2	Yes
2	46-328000G979	823	MSM1-J10 TO MR2-A11-J24	No
3	46-317359G975	824TR	PP1-J10 TO FJ1	Yes
4	46-317359G977	825TR	PP1-J48 TO MSM1-A1-J8	Yes
5	46-328000G978	826	FJ3 TO MSM1-A1-J9	No
6	46-328578P1	827	FJ3 TO FJ4 & MS5A5	Yes
7	46-328000G974	828	MS1-A3-A1-J403 TO PP1-J10	Yes
8	46-328000G976	829	MS1-FJ2 TO PP1-J48	No
9a	2279237	830	Magnet Room Interface 15 pin D (18-ft) RUO Preamp 9 pin D (15-ft) Pressure Transducer 2 cond pair (9-ft) Reed Switch 10 pin Bendix (18-ft) coldhead <i>Used on HFO (2 coldheads)</i>	No
9b	2222797	830	Magnet Room Interface 15 pin D (18-ft) RUO Preamp 9 pin D (15-ft) Pressure Transducer 2 cond pair (9-ft) Reed Switch <i>Used on Core LCC Cable Kits</i>	Yes
9c	2222797-2	830	Magnet Room Interface 15 pin D (6-ft) RUO Preamp 9 pin D (9-ft) Pressure Transducer 2 cond pair (40-ft) Reed Switch <i>Used on 7.0T Cable Kits</i>	No
9d	2108726	830	2 cond pair for Reed Switch 9 pin D for Pressure Transducer <i>Used on S-Series and CX Magnets</i>	Yes
10	46-328000G980	831	FJ1 TO MSM1-A1-J7	Yes
11a	2214062	832	MS1-A2-J1 to MS1 Sleeve and MS1 Coldhead Cable to coldhead is 2-ft Cable to sleeve is 2-ft. <i>Used on LCC and LCC300</i>	Yes
11b	2214062-2	832	MS1-A2-J1 to MS1 Sleeve and MS1 Coldhead Cable to coldhead is 6-ft Cable to shield sensors near vacuum port is 30-ft. <i>Used on 7.0T</i>	No
11c	2271499	832	Coldhead Head Adapter Cable, Shielded, cable length is 2-ft. <i>Used on 0.7T HFO</i>	Yes
12	46-328000G981	833	FJ4 TO MS5-A1-A6-JR	Yes
13	2263158	913	PP1-J62 to MS1-A1-A4 P302-3	No
14	2262699	914	Run 916-FJ5 to PP1-J62	Yes

Item	Part Number	Run #	Description	FRU
15	2263160	915	MSM1A2-J1 TO FJ6	Yes
16	2263159	916	MSM1-J7 TO FJ1, FJ5 AND FJ6	No
17	2264859	919	PP1-J10 to MS1-A1-A4 P302-1	No
18	2276568	940	MSM3-RS-232 TO MSM1-J1	Yes
19	2271497	942	MSM1-J5 to customer network jack	Yes
20	2404555	1278	Run 827-FJ4 to A1JR	No
21	46-271600G37	1301	MSM1-J8 to PP3-J9	No
22	46-271600G37	1302	Run 1307-FJ7 to PP3-J10	No
23	2404562	1307	MSM1-J12 to FJ3b and FJ7	No
24	2404554	1308	PP3-J10 to MS1 A3 – J4	No
25	2404553	1309	MSM1-J9 to FJ3a and Lock Coil Timer	No
26	46-271609G37	1310	Lock Coil Timer to PP3-J2	No
27	2404561	1311	PP3-J2 to Mag Instrument Box – Lock 1	No
28	46-271601G37	1312	Run 916 – FJ5 to PP3-J11	No
29	2404558	1313	PP3-J11 to Mag Instrument Box – LHe 1	No
30	5183297	1410	MS1 A2-J1 to MS1 A1-J6 and MS1 A4-J2 and MS1 A4-J3	No
31	2255874	N/a	CABLE, RJ11 MODULAR,6-COND. 7FT	No
32	2216697	N/a	RJ11 CABLE (25 FEET) – phone wire	Yes
33	2204485	N/a	System Cabinet Interface Cable (LX systems)	Yes

Table 11-2: Component Parts

Item	Part Number	Description	FRU
1	2394952	Advantech - Magnet Monitor 3	Yes
2	2219341	RUO Pre-amp	Yes
3	2333825	Lake Monitor - Water Flow (Turbine)/Temp Sensor Assembly	Yes
4	5113506	Lake Monitor - Water Flow Meter (Turbine) Rebuild Kit	Yes
5	2299843-2	SUNX - Pressure Transducer (Used on 0.5T, 1.0T, 1.5T, 3.0T and 7.0T Cylindrical Magnets)	Yes
6	2299843	SUNX – Pressure Transducer (Used on 0.7T HFO)	Yes
7	2369032	RF Shield Cover for SUNX Pressure Transducer	Yes
8	2108790	Reed Switch	Yes
9	2266628	Remote Alarm Box	Yes
10	5180301	Remote Alarm Kit	Yes
11	2276094	UPS	Yes
12	46-265067P1	I/F Panel Cable Screw Lock Connector	Yes
13	2404712	Lock Coil Timer (used on 7.0T)	No
14	2245773	Multi-Tech Global Modem	Yes

15	2249450	Global Modem Adapter Assembly for Europe	Yes
16	46-328475P1	4 Line Multiplexer w/115 vac ps	
	46-328475P3	4 Line Multiplexer w/220 vac ps	
	46-328475P2	8 Line Multiplexer w/115 vac ps	
	46-328475P4	8 Line Multiplexer w/220 vac ps	
17	2385399	Satellite Modem Kit	Yes
18	46-252294P1	Swagelok ¼-inch Union Tee	Yes
19	46-296816P1	Adapter, BNC Tee, 50 ohm	Yes

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Chapter 12 Appendix

1 Appendix

1.1 Prerequisite Information for Non-Proprietary Operation

To configure Magmon3 network connectivity, the user must obtain information from the customer's network system administrator. This section details the type of information required based on connectivity type and system type. Make sure to start this information-gathering process at least 3 weeks in advance of the physical installation.

1.1.1 For Non-HFO Systems (Fixed Sites and Mobiles)

Neither a Network Connection or Cryogen System ID is used.

- **Magnet Type (Required):** S2, S3, S4, S5, SX, CX, LCC, LCC300, 7.0T, VMX, MAX, Compact, Other.
- **Compressor Type (Required):** Leybold or Sumitomo.

1.1.2 For HFO Systems

Network Cable connection from Magmon3 J5 to OW – Host Lan Hub J7.

- **Magnet Type (Required):** HFO
- **Compressor Type (Required):** Sumitomo.

DHCP is not used because of the coldhead cycling hardware and communication requirement. Must use the following:

- **Static Internet Protocol (IP) Address:** Obtain from customer IT person.
- **Subnet Address:** Obtain from customer's IT person.
- **Gateway Address:** Obtain from customer's IT person.
- **Domain Name Service (DNS):** Obtain from customer's IT person.

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Chapter 13 Vendor Documents

1 Vendor Documents

1.1 MultiTech - Global Modem Manual

Printable version is available as a .pdf file or found on the MultiTech website: <http://www.multitech.com/DOCUMENTS/Families/MultiModemZBA/manuals.asp>.

1.2 SUNX DP2-41N Instruction Manual

Printable version is available as a .pdf file or found on the SUNX website: <http://pewa.panasonic.com/acsd/sunx-sensors/> . Once the webpage displays, type **DP2-41N** in the *Search* field and press [Go].

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